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DRDO-RDEE-MBS-IPL-032-2024

**ILLUSTRATED SPARES PART LIST
(ISPL)
(VOLUME I)
FOR
MODULAR BRIDGE**

25/10/2024

R&DE(E)



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**Government of India, Ministry of Defence,
Defence R&D Organization,
Research and Development Establishment (Engineers)
Kalas, Alandi Road, Dighi PO,
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Reviewed By:	Amit Manglik, Sc 'F'		
Approved and Issued By:	DR. Rahul Harshe, Sc 'F' Group Head, MBG		

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(VOLUME I)
FOR
MODULAR BRIDGE**

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(VOLUME I)
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Date :

Name : Amit Manglik

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WARNING

1. System to be SWITCH OFF before servicing the equipment.
2. In event of unusual operation (unusual sounds, oil leakage, smoke etc.) immediately stop operation and take appropriate corrective measures.
3. Operating sequence given in the 'User Handbook' must be strictly followed.
4. Person standing near bridge / vehicle must maintain safe distance from the bridge / vehicle when it is in operation.
5. No loose component, tools must be kept on bridge during bridging operation mode.
6. For repair on the critical assemblies / components / parts, the Original Equipment Manufacturer (OEM) must be consulted.
7. Before movement of the launcher vehicle and carrier vehicle, check transportation locks.
8. Before movement of launcher vehicle, make sure that PTO is disengaged.

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**DEMOLITION OF EQUIPMENT
TO PREVENT ENEMY USE**

THE DEMOLITION OF THE EQUIPMENT WILL BE HELD ONLY ON THE ORDER OF THE
COMMANDING OFFICER.

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FIRST AID IN CASE OF ELECTRICAL SHOCK

1.	Shift the person to the nearest medical centre immediately, who is affected by electrical shock.
2.	i) Terminate the battery connection of vehicle. ii) Find fault in defective cable / wire. iii) Replace defective cable / wire from vehicle chassis.

FIRST AID IN CASE OF ELECTRIC SHOCK

Figure. 1

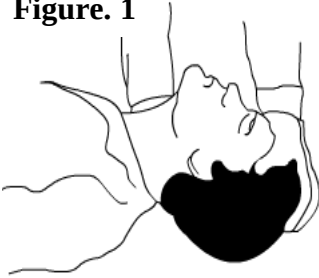


Figure. 2



Figure. 3



Figure. 4



- | | |
|----|--|
| 1. | SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME. |
| 2. | (a) Place the victim in the supine position.

(b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat. (Figure 1) |
| 3. | If the jaw is rigid, try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages thus cleared, tilt the head backward and force the jaw forward from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages. (Figure 2-3) |
| 4. | (a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other. (Figure 3)

(b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG). (Figure 4) |
| 5. | When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre. |

**HAVE SOMEONE ELSE AND SEND FOR A DOCTOR.
KEEP PATIENT WARM AND LOOSEN HIS CLOTHING.
DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS
CONSCIOUS.**

ABBREVIATIONS

Sr. No.	Abbreviation Issued	For
1	APU	Auxiliary Power Unit
2	BCM	Bridge Centre Module
3	BEM	Bridge End Module
4	BSB	Bank Seat Beam
5	CV	Carrier Vehicle
6	DCV	Directional Control Valve
7	DPT	Dye Penetration Test
8	ECP	Electronic Control Panel
9	FL	Front Left
10	FR	Front Right
11	GVW	Gross Vehicle Weight
12	HMV	High Mobility Vehicle
13	HPP	Hydraulic Power Pack
14	HSLA	High Strength Low Alloy
15	JC	Joystick Console
16	LP	Launching Platform
17	LED	Light Emitting Diode
18	LS	Launching System
19	LVJB	Launcher Vehicle Junction Box
20	MB	Modular Bridge
21	MDP	Main Distribution Panel
22	NAS	National Aerospace Standard
23	NCM	Nose Centre Module
24	NEM	Nose End Module

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Sr. No.	Abbreviation Issued	For
25	NPC	Nose Pushing Cylinder
26	MBS	Modular Bridge System
27	MHD	Material Handling Device
28	MLC	Military Load Classification
29	MMI	Man Machine Interface
30	OEM	Original Equipment Manufacturer
31	PDU	Power Distribution Unit
32	PTO	Power Take Off
33	PVG	Proportional Valve Group
34	QDC	Quick Disconnect Coupling
35	RL	Rear Left
36	RR	Rear Right
37	SF	Sub Frame
38	SP	Sliding Platform

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PREFACE

1. This Illustrated Spare Parts List has been prepared for the purpose of identifying any component, part or assembly of parts likely to be required as replacements. All concerned will use the Cat Sec. / Part No., NSN and DCAN and designations given in this list when demanding spare part or when referring in correspondence to any of the equipment. All demands for items listed herein must be submitted to appropriate authority in accordance with standing instruction.
2. The Cat Sec. / Part No. (when quoted initially) and Designation of all 'P' marked items quoted in the list will be supercede by NSN and codified designation respectively as soon as such items will be codified.
3. When demanding spare parts the following particulars must be quoted:
 - (a). Relevant Section / Sub-section of Catalogue of Ordnance Stores (i.e. COS) or Catalogue of Engineer Stores (i.e. COES) or Catalogue of Mechanical Transport Stores (i.e. CMTS) or Catalogue of Clothing (i.e. COC).
 - (NOTE - The above mentioned Catalogues belong to Army only. Navy and Air Force to mention their respective Catalogues under which the equipment/items are provided by them)
 - (b). Cat Sec. / Part No. or NSN or DCAN of spare parts along with their designation.
 - (c). Designation and Cat Sec. / Part No. or NSN or DCAN of the main equipment.
4. Certain items included in this ISPL have not been marked 'P' (Provided) by concerned MAG since they are not considered to be provisioned for maintenance of the equipment. But User Unit / EME workshop may demand certain non 'P' items which they consider essential for maintenance purpose.
5. All amendment to this list will be forwarded to all concerned by the concerned AHSP / department primarily responsible for providing the equipment. It is essential that this list should be carefully amended and date of each amendment entered.
6. Queries relating to this ISPL should be referred to Controller / HOE of the concerned office by whom this ISPL has been published. The name and address of the publisher is shown on cover page / cover leaf of this ISPL.

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SECTION - A

MODULAR BRIDGE SYSTEM

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PLATE - I



MODULAR BRIDGE SYSTEM - VIEW 1

RESTRICTED

RESTRICTED

PLATE - I

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
1.	01			GA- LAUNCHING SYSTEM	2	410A100000000			
2.	02			GA- BRIDGE SUPER STRUCTURE	1	410A200000000			
3.	03			GA- NOSE SUPER STRUCTURE	1	410A300000000			
4.	04			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 1	1	410A400000000			
5.	05			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 2	1	410A500000000			
6.	06			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 3 & 4	2	410A600000000			
7.	07			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 5	1	410A700000000			
8.	08			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 6	1	410A800000000			
9.	09			CARRIER VEHICLE (CV) STOWAGE ASSEMBLY 7	1	410A900000000			

MODULAR BRIDGE SYSTEM - VIEW 1

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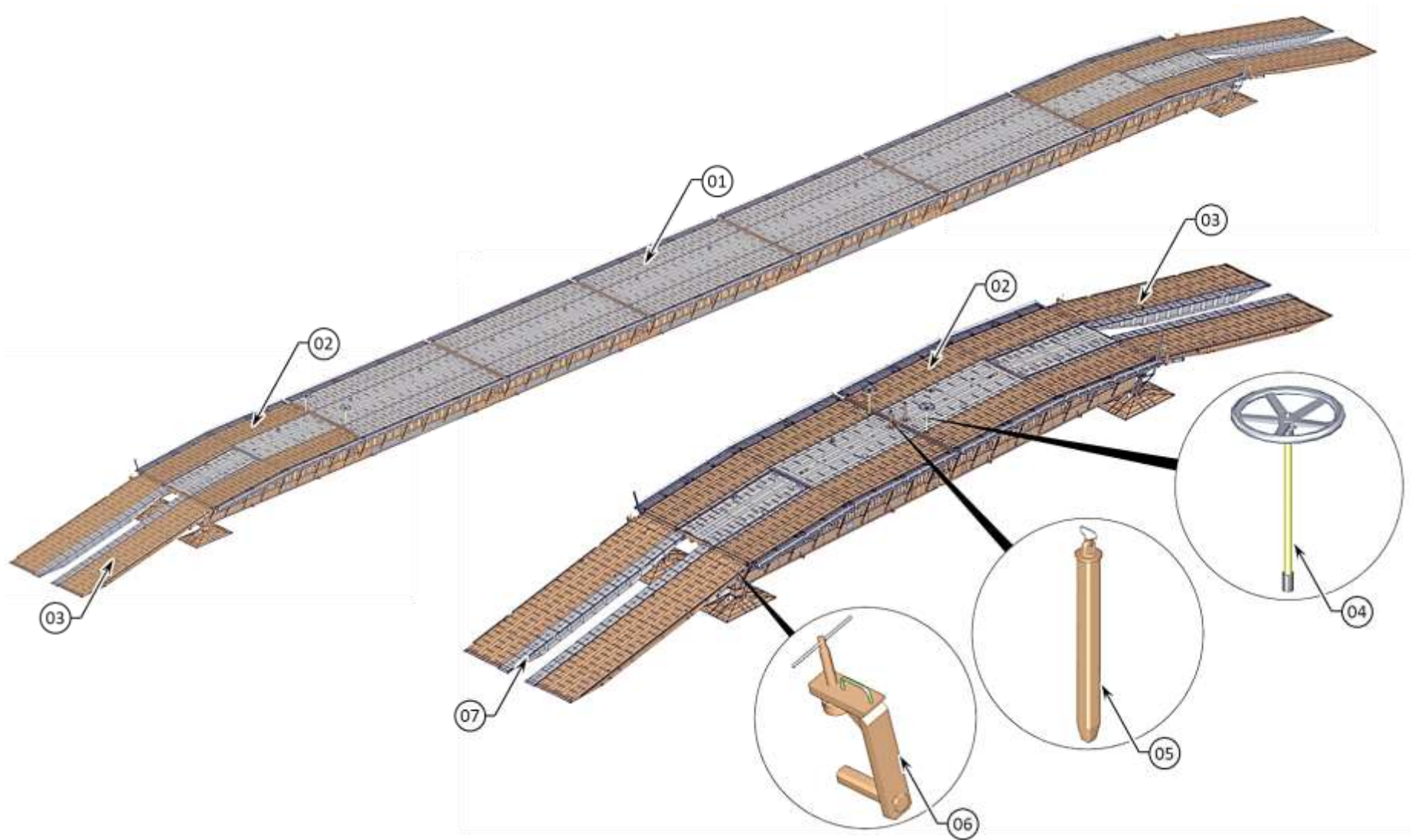
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SECTION - B

GENERAL ASSEMBLY OF BRIDGE SUPERSTRUCTURE

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PLATE - I



GENERAL ASSEMBLY OF BRIDGE SUPERSTRUCTURE

RESTRICTED

RESTRICTED

PLATE - I

SR. NO.	PLATE REF / ITEM NO.	'P' PRO VIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
1.	01			BRIDGE CENTRE MODULE ASSEMBLY	5	410A210000000		2,01,00,126	
2.	02			BRIDGE END MODULE ASSY	4	410A220000000		2,41,77,017	
3.	03			RAMP ASSY	8	410A230000000		13,32,265	
4.	04			HANDWHEEL ASSY	4	410A240000000		59,870	
5.	05			BRIDGE TO NOSE LOCKING PIN ASSY	4	410A250000000		23,180	
6.	06			RAMP LOCK ARRANGEMENT	8	410A260000000		59,560	
7.	07			RAMP FILLER PANEL ASSY	8	410A290000000		1,99,874	

Note: 1 set of Modular Bridge (MB) comprises of above sub-assemblies of bridge superstructure.

GENERAL ASSEMBLY OF BRIDGE SUPERSTRUCTURE

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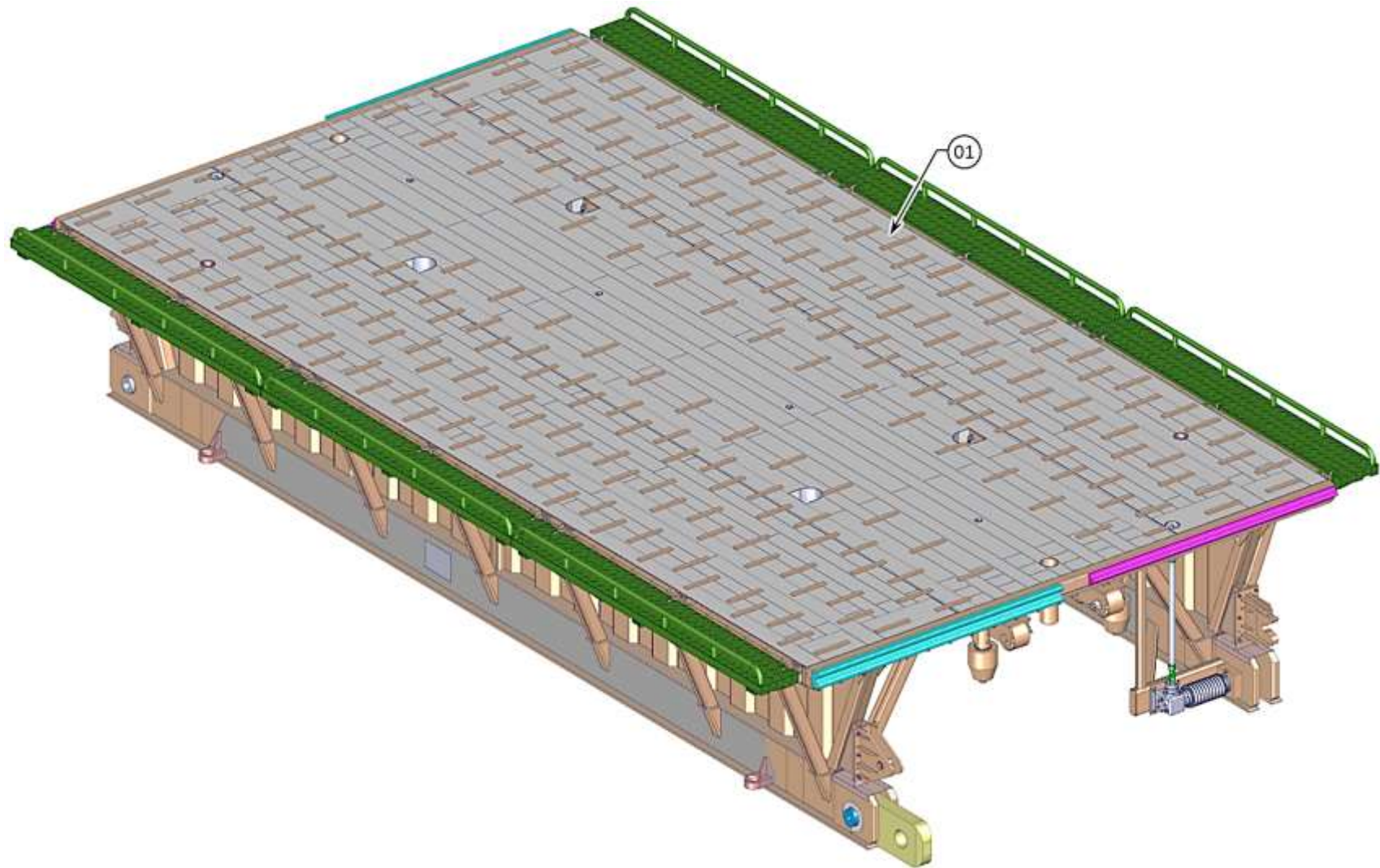
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SUB - SECTION - 1
BRIDGE CENTRE MODULE ASSEMBLY

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PLATE - II



BRIDGE CENTRE MODULE ASSEMBLY

RESTRICTED

PLATE - II

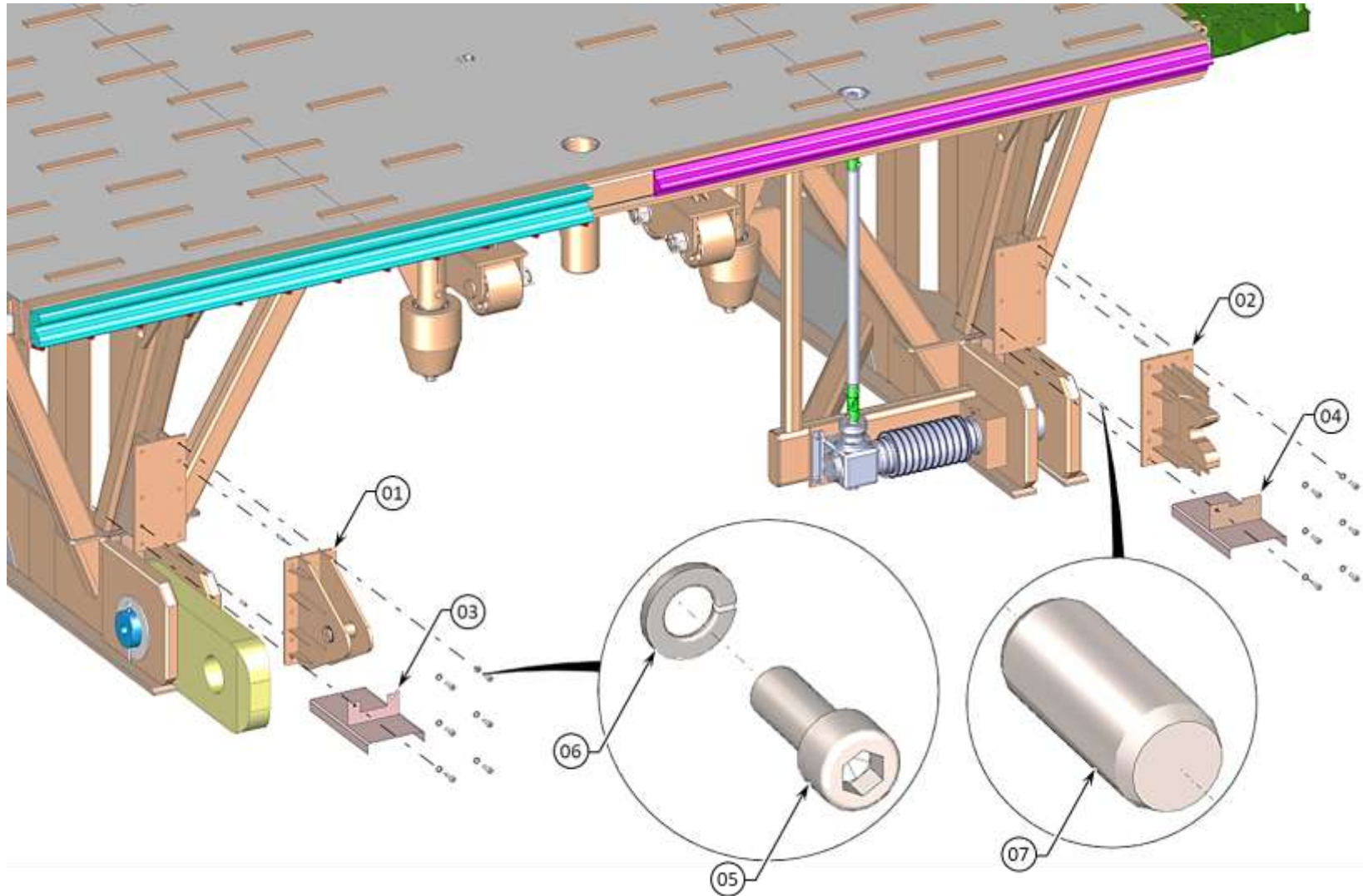
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
1.	01			BRIDGE CENTRE MODULE ASSEMBLY	1	410A210000000		2,01,00,126	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY

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PLATE - III



BRIDGE CENTRE MODULE ASSEMBLY - VIEW 1

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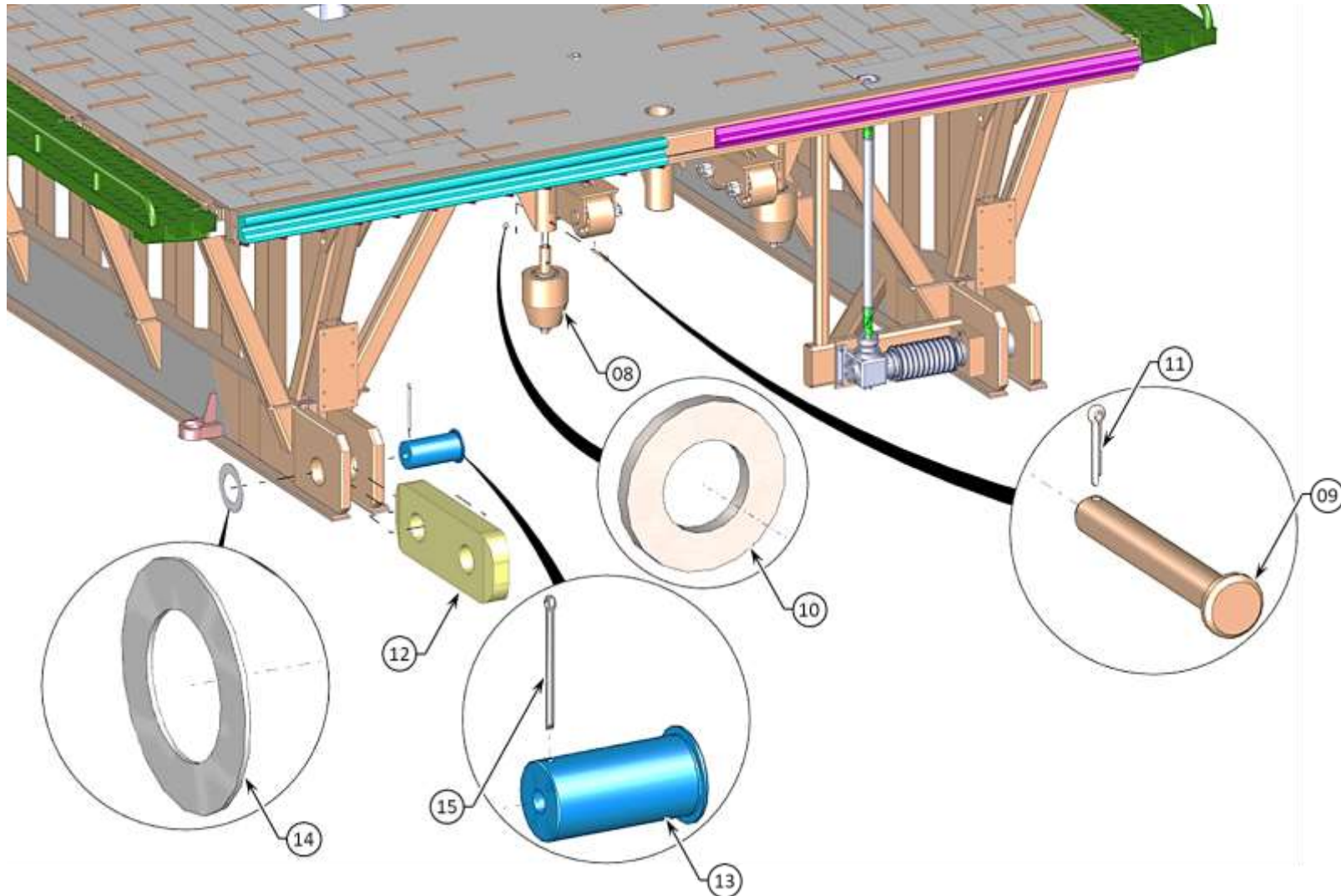
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PLATE - III

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
2.	01			GUIDE BEAK MALE ASSEMBLY	2	410A213000000		84,557	
3.	02			GUIDE BEAK FEMALE ASSY	2	410A214000000		50,978	
4.	03			COVER ASSY-1	2	410A21A000000		2,988	
5.	04			COVER ASSY-2	2	410A21B000000		2,988	
6.	05			HEX. SOCKET HEAD CAP SCREW ISO 4762-M10x25-12.9	24	410A210001700		26	
7.	06			SPRING LOCK WASHER B10-IS 3063 - STEEL	24	410A210002000		18	
8.	07			PARALLEL PIN IS 2393 10h8 X 26 - St	8	410A210002500		27	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY - VIEW 1

RESTRICTED
PLATE - III (CONTINUED)



BRIDGE CENTRE MODULE ASSEMBLY - VIEW 2

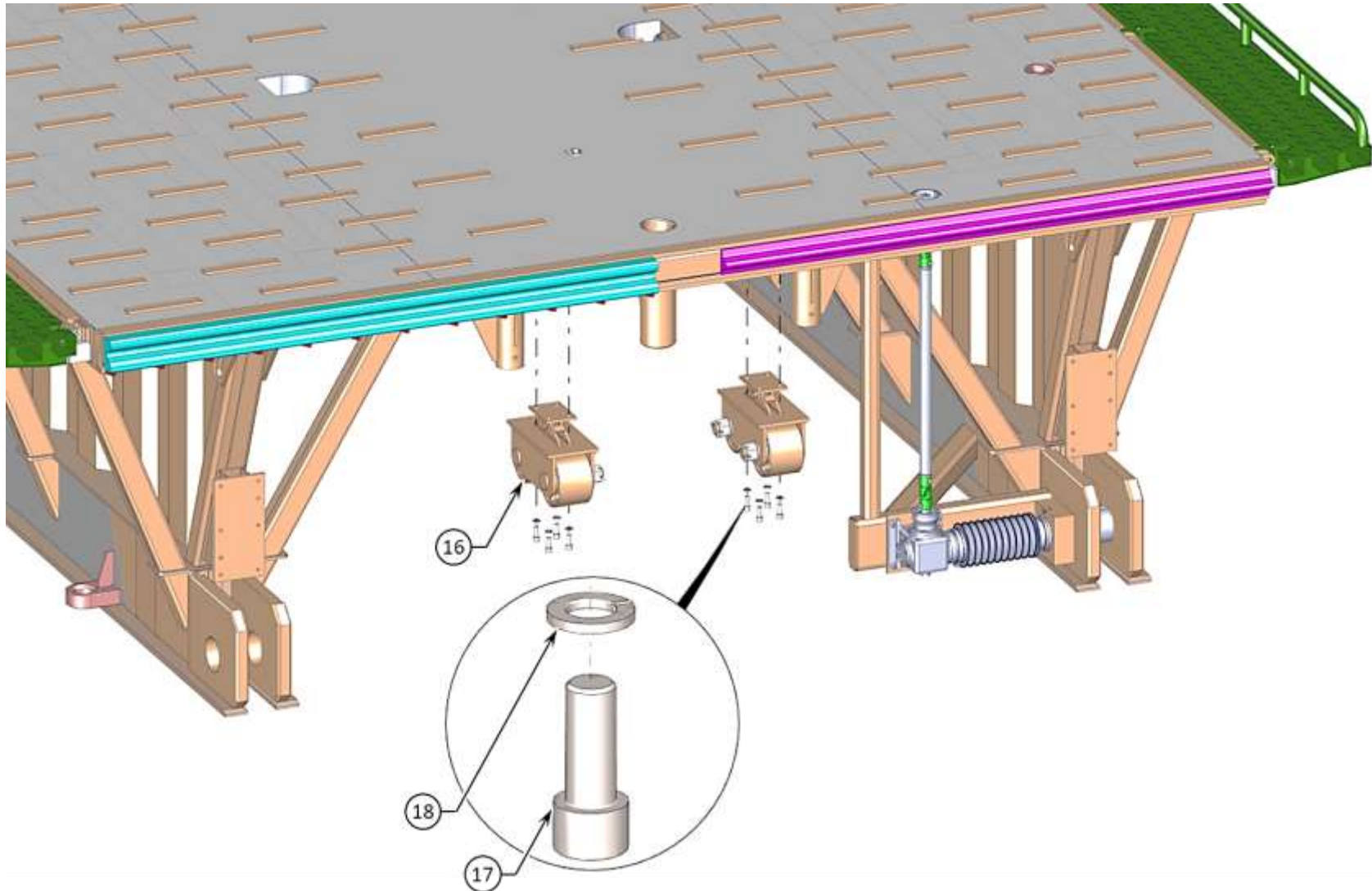
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PLATE - III (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
9.	08			GUIDE ROLLER ASSEMBLY	4	410A217000000		2,16,140	
10.	09			LOCKING PIN FOR GUIDE ROLLER ASSY	4	410A210000700		35,766	
11.	10			MACHINED WASHER 13 - IS:2016 STEEL	4	410A210001900		27	
12.	11			SPLIT PIN IS 549-3.2 X 16-St	4	410A210002200		9	
13.	12			LINK PLATE	2	410A210000400		1,43,683	
14.	13			BRIDGE LOCKING PIN	2	410A210000600		1,07,324	
15.	14			SPACER	2	410A210001000		2,742	
16.	15			SPLIT PIN IS 549-10 X 125-St	2	410A210002300		89	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY - VIEW 2

RESTRICTED
PLATE - III (CONTINUED)



BRIDGE CENTRE MODULE ASSEMBLY - VIEW 3

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PLATE - III (CONTINUED)

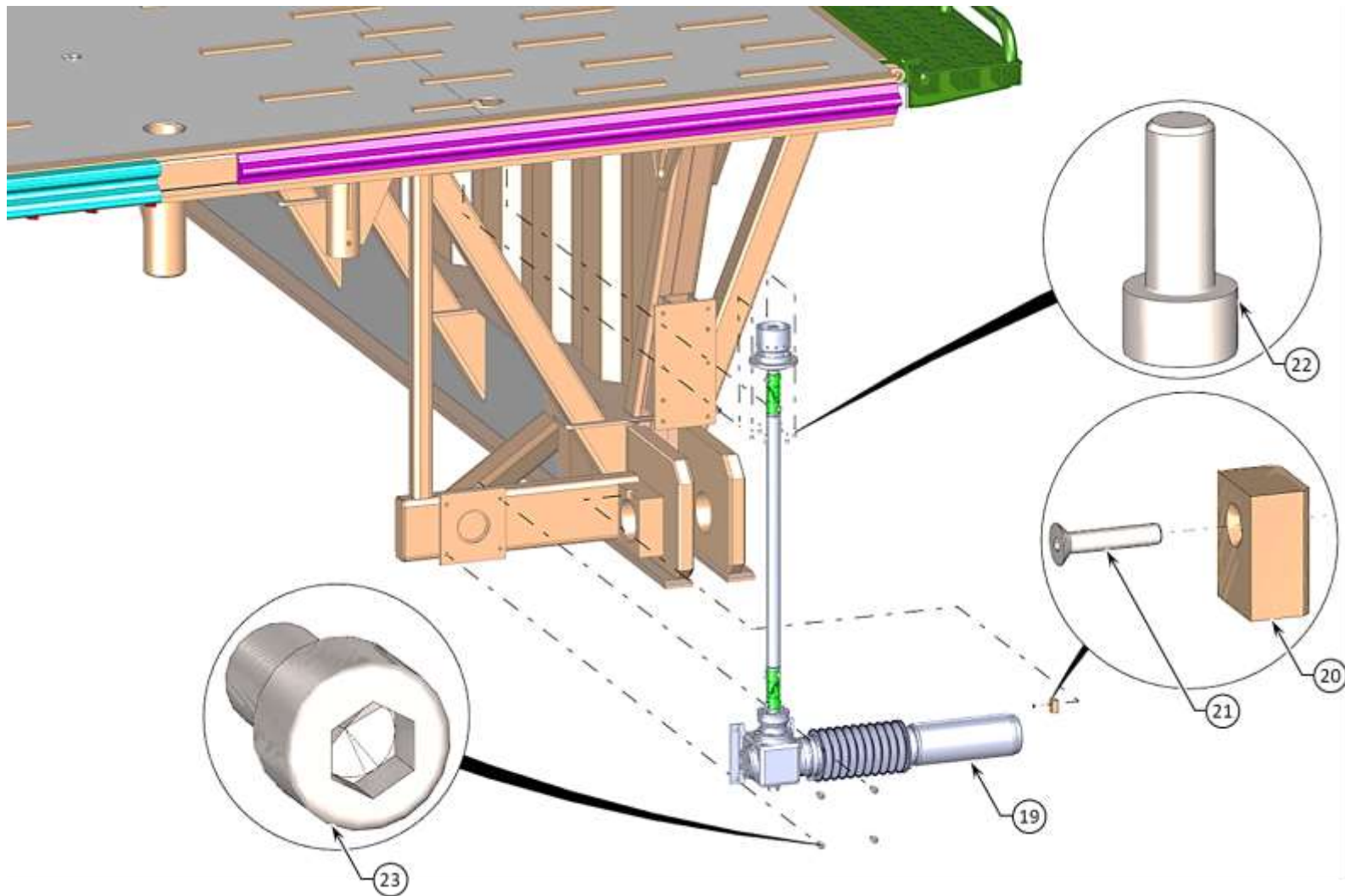
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
17.	16			SUPPORT ROLLER ASSEMBLY	4	410A215000000		3,34,100	
18.	17			HEX. SOCKET HEAD CAP SCREW ISO 4762-M12x30-12.9	16	410A210001800		40	
19.	18			SPRING LOCK WASHER B12-IS 3063 - STEEL	16	410A210002100		18	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY - VIEW 3

RESTRICTED

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PLATE - III (CONTINUED)



BRIDGE CENTRE MODULE ASSEMBLY - VIEW 4

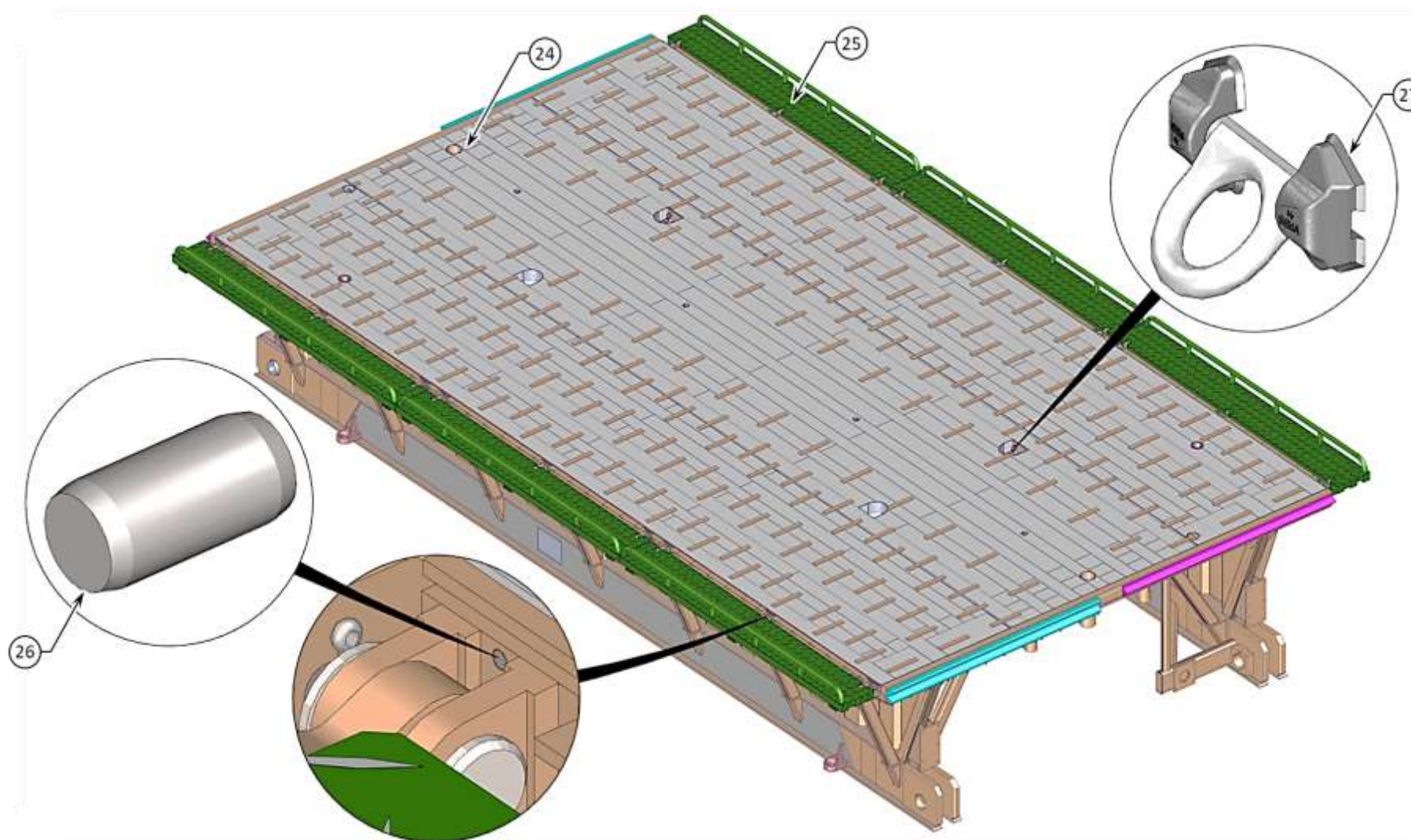
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PLATE - III (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
20.	19			BRIDGE TO BRIDGE LOCKING MECHANISM	2	410A218000000		5,93,759	
21.	20			TENON	2	410A210000900		2,376	
22.	21			HEXAGON SOCKET COUNTERSUNK HEAD SCREW M5x25 IS 6761	2	410A210001400		9	
23.	22			HEX. SOCKET HEAD CAP SCREW ISO 4762-M6x16-12.9	12	410A210001500		18	
24.	23			HEX. SOCKET HEAD CAP SCREW ISO 4762-M10x16-12.9	8	410A210001600		27	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY - VIEW 4

RESTRICTED
PLATE - III (CONTINUED)



BRIDGE CENTRE MODULE ASSEMBLY - VIEW 5

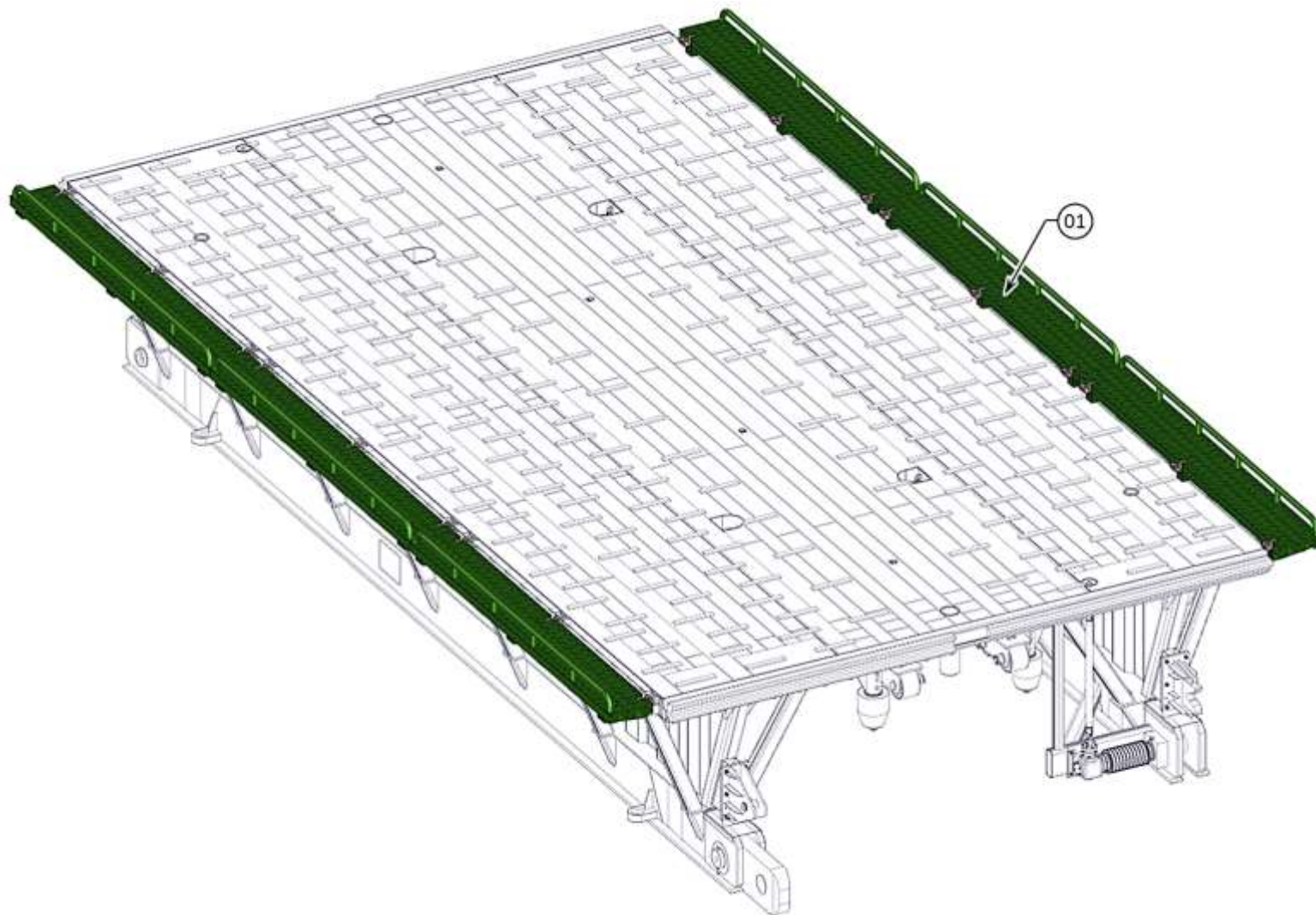
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PLATE - III (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
25.	24			BRIDGE CENTRE MODULE STRUCTURE	1	410A211000000		1,09,90,529	
26.	25	P		CURB ASSEMBLY	6	410A212000000		2,52,120	
27.	26			PARALLEL PIN IS 2393 6m6 X 12 - St	36	410A210002400		18	
28.	27			LIFTING HOOK - MAKE:RUD, TYPE:4T	4	410A210002600		5,666	

Note: The quantity mentioned in above table is for 1 BCM. 1 set of MB contains 5 nos. of BCM.

BRIDGE CENTRE MODULE ASSEMBLY - VIEW 5

RESTRICTED
PLATE - IV



CURB ASSY (BCM)

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PLATE - IV

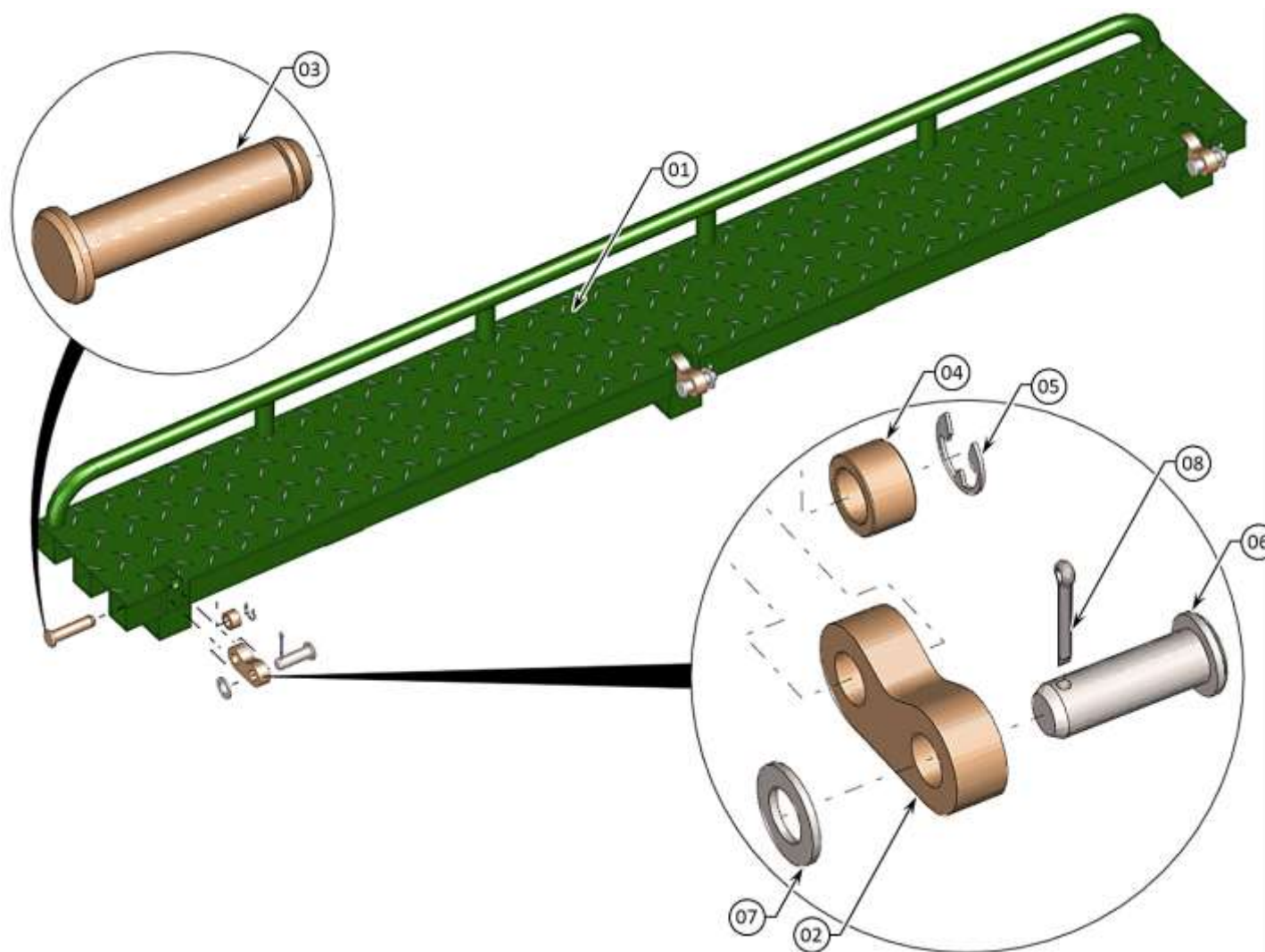
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
29.	01	P		CURB ASSEMBLY (BCM)	1	410A212000000		2,52,120	

Note: The quantity mentioned in above table is for 1 curb assy (1 BCM contains 6 nos. of curb assy). 1 set of MB contains 5 nos. of BCM.

CURB ASSY (BCM)

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PLATE - V



CURB ASSY (BCM) - VIEW 1

RESTRICTED

PLATE - V

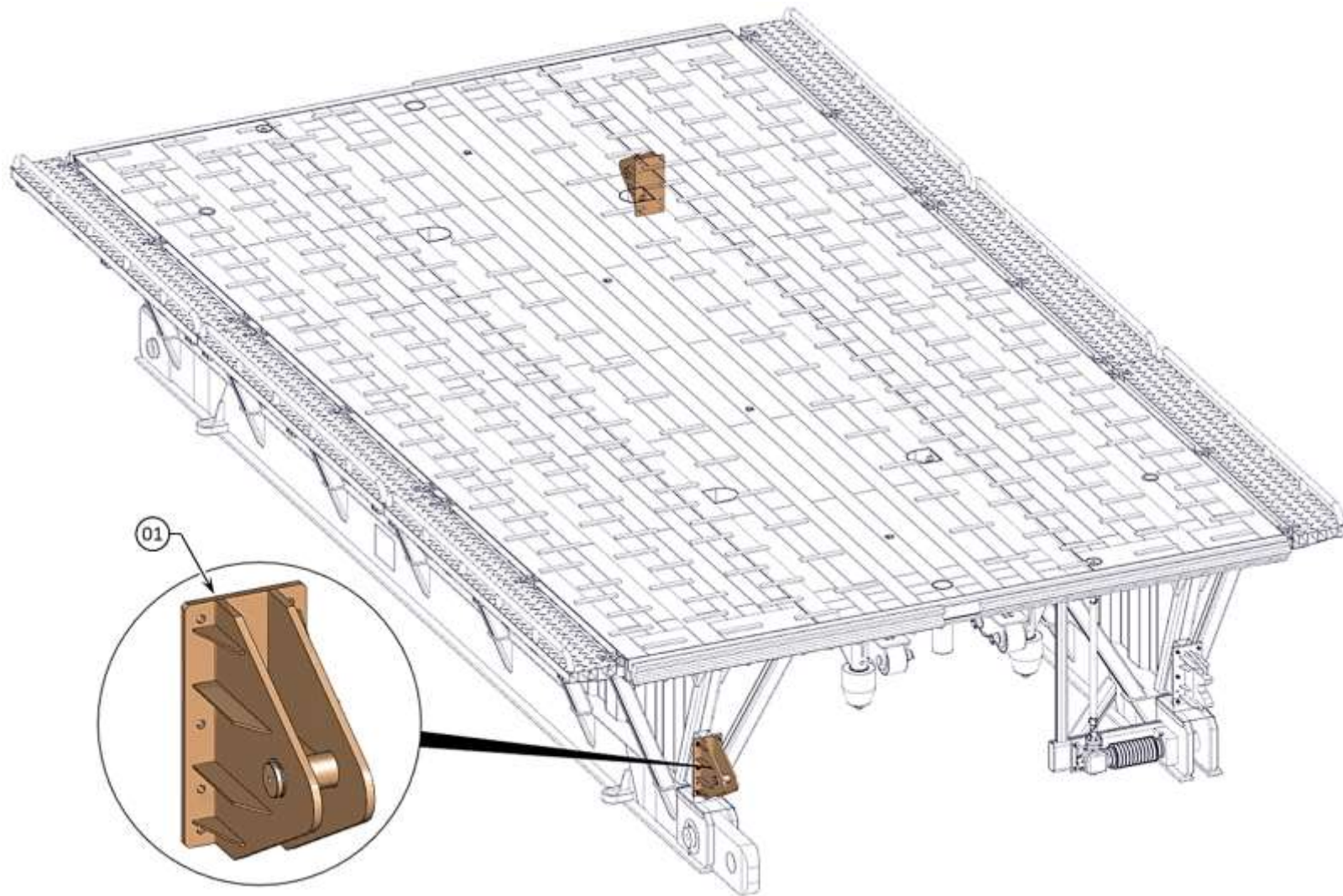
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
30.	01			CURB ASSY STRUCTURE	1	410A212100000		1,20,437	
31.	02			LINK PLATE	3	410A212000100		2,742	
32.	03			PIN	3	410A212000200		17,536	
33.	04			BUSH	3	410A212000300		17,426	
34.	05			CIRCLIP 15 x 2N IS:3075 (PART-3)	3	410A212000400		18	
35.	06			CLEVIS PIN ISO 2341-B-20x60x5-St	3	410A212000500		146	
36.	07			MACHINED WASHER 21-IS:2016-STEEL	3	410A212000600		9	
37.	08			SPLIT PIN IS 549-5x32-St	3	410A212000700		9	

Note: The quantity mentioned in above table is for 1 curb assy (1 BCM contains 6 nos. of curb assy). 1 set of MB contains 5 nos. of BCM.

CURB ASSY (BCM) - VIEW 1

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PLATE - VI



GUIDE BEAK MALE ASSEMBLY (BCM)

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PLATE - VI

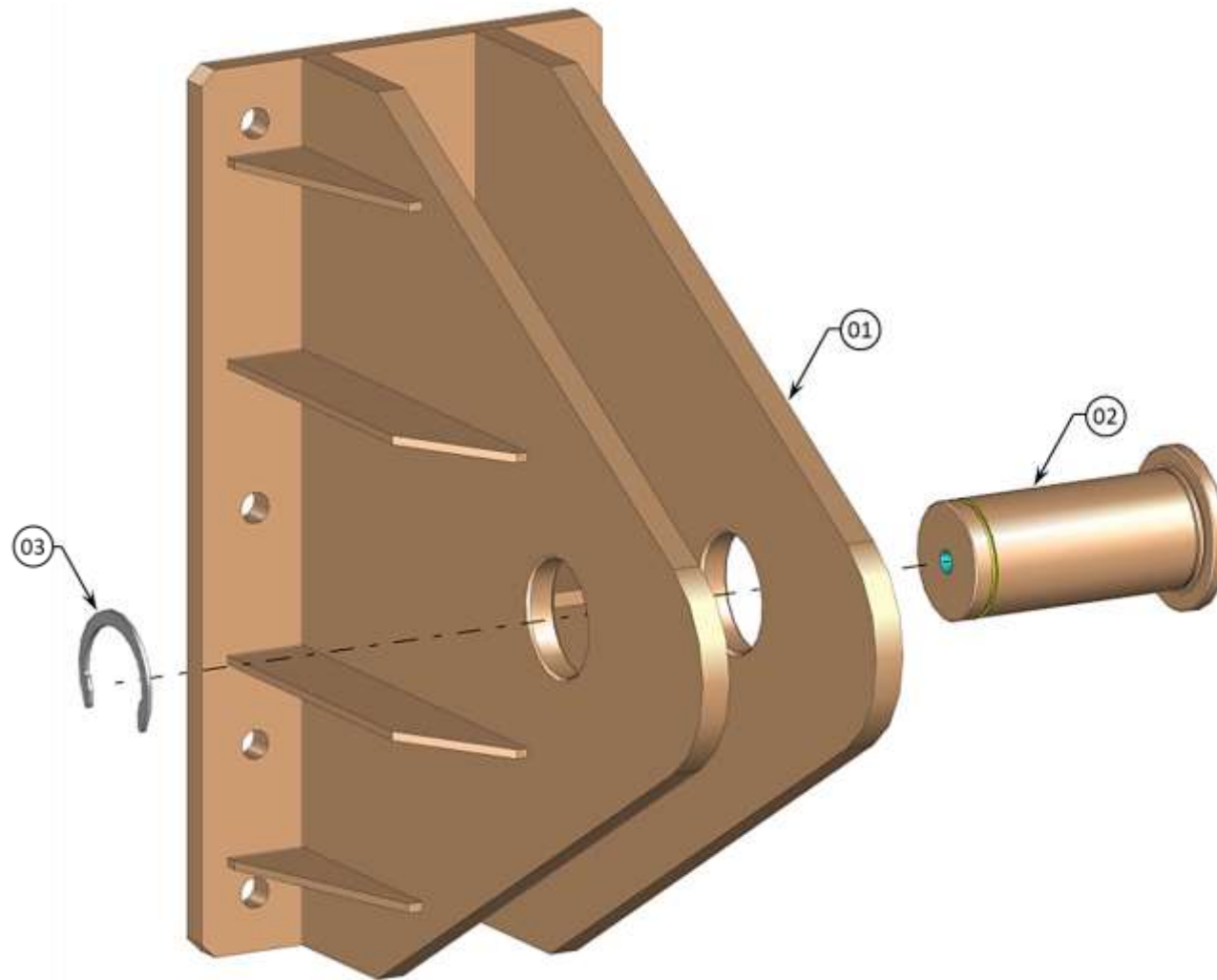
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
38.	01			GUIDE BEAK MALE ASSEMBLY (BCM)	1	410A213000000		84,557	

Note: The quantity mentioned in above table is for 1 guide beak male assembly (1 BCM contains 2 nos. of guide beak male assembly). 1 set of MB contains 5 nos. of BCM.

GUIDE BEAK MALE ASSEMBLY (BCM)

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PLATE - VII



GUIDE BEAK MALE ASSEMBLY (BCM) - VIEW 1

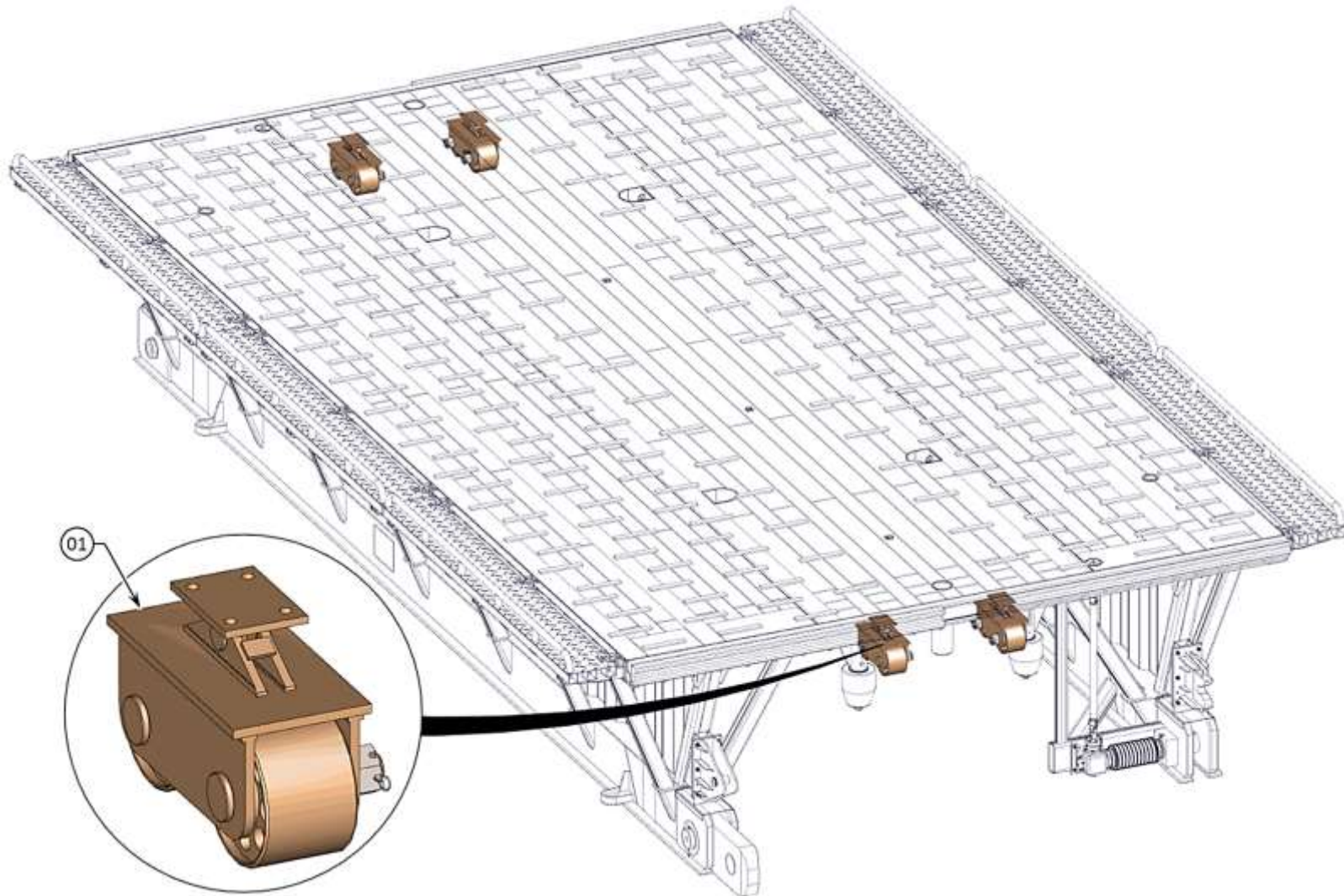
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PLATE - VII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
39.	01			GUIDE BEAK MALE BRACKET	1	410A213100000		55,102	
40.	02			PIN FOR GUIDE BEAK MALE	1	410A213000100		17,151	
41.	03			CIRCLIP 40 x 1.75 N IS:3075 (PART 1)	1	410A213000200		18	

Note: The quantity mentioned in above table is for 1 guide beak male assembly (1 BCM contains 2 nos. of guide beak male assembly). 1 set of MB contains 5 nos. of BCM.

GUIDE BEAK MALE ASSEMBLY (BCM) - VIEW 1

RESTRICTED
PLATE - VIII



SUPPORT ROLLER ASSEMBLY (BCM)

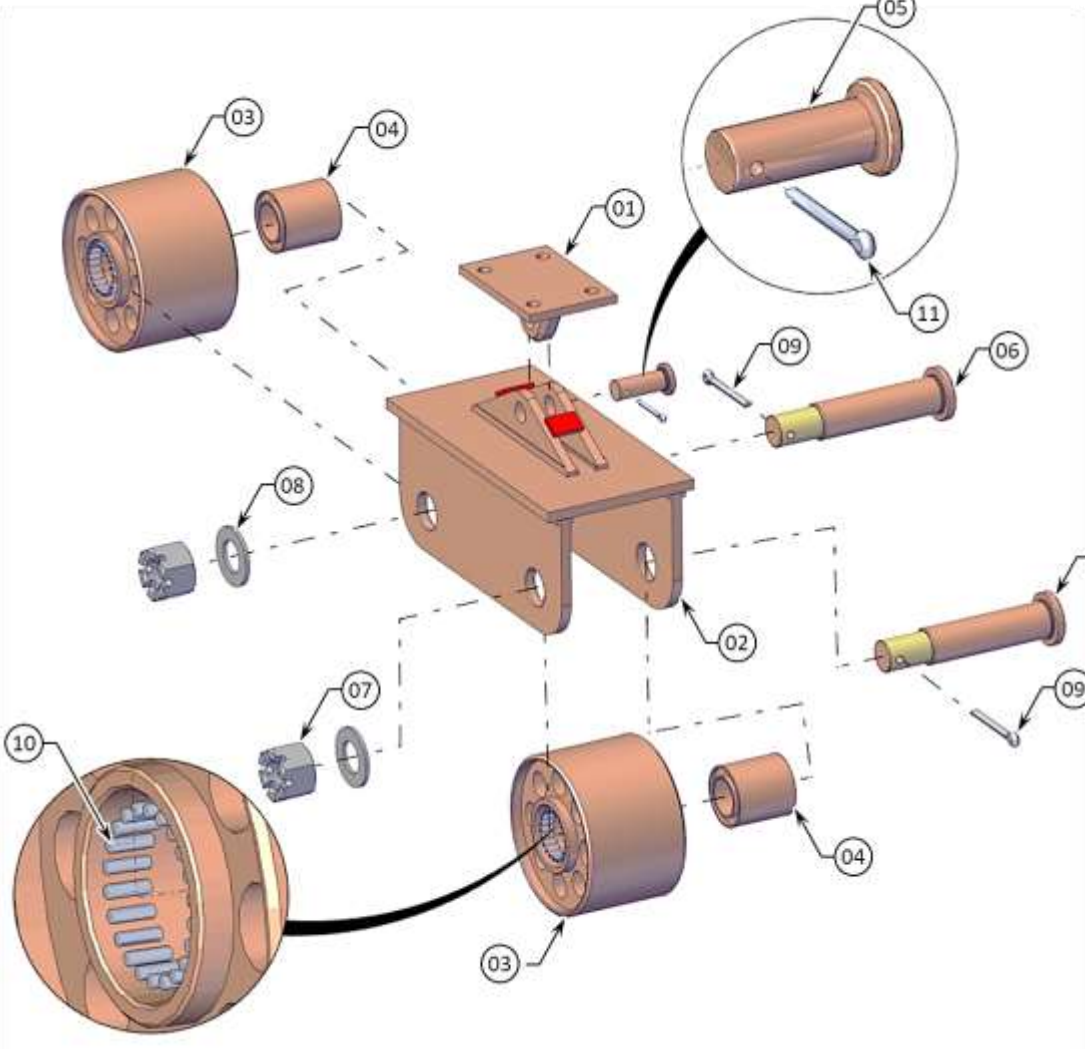
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PLATE - VIII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
42.	01			SUPPORT ROLLER ASSEMBLY (BCM)	1	410A215000000		3,94,294	

Note: The quantity mentioned in above table is for 1 support roller assembly (1 BCM contains 4 nos. of support roller assembly). 1 set of MB contains 5 nos. of BCM.

SUPPORT ROLLER ASSEMBLY (BCM)

PLATE - IX



SUPPORT ROLLER ASSEMBLY (BCM) - VIEW 1

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RESTRICTED

PLATE - IX

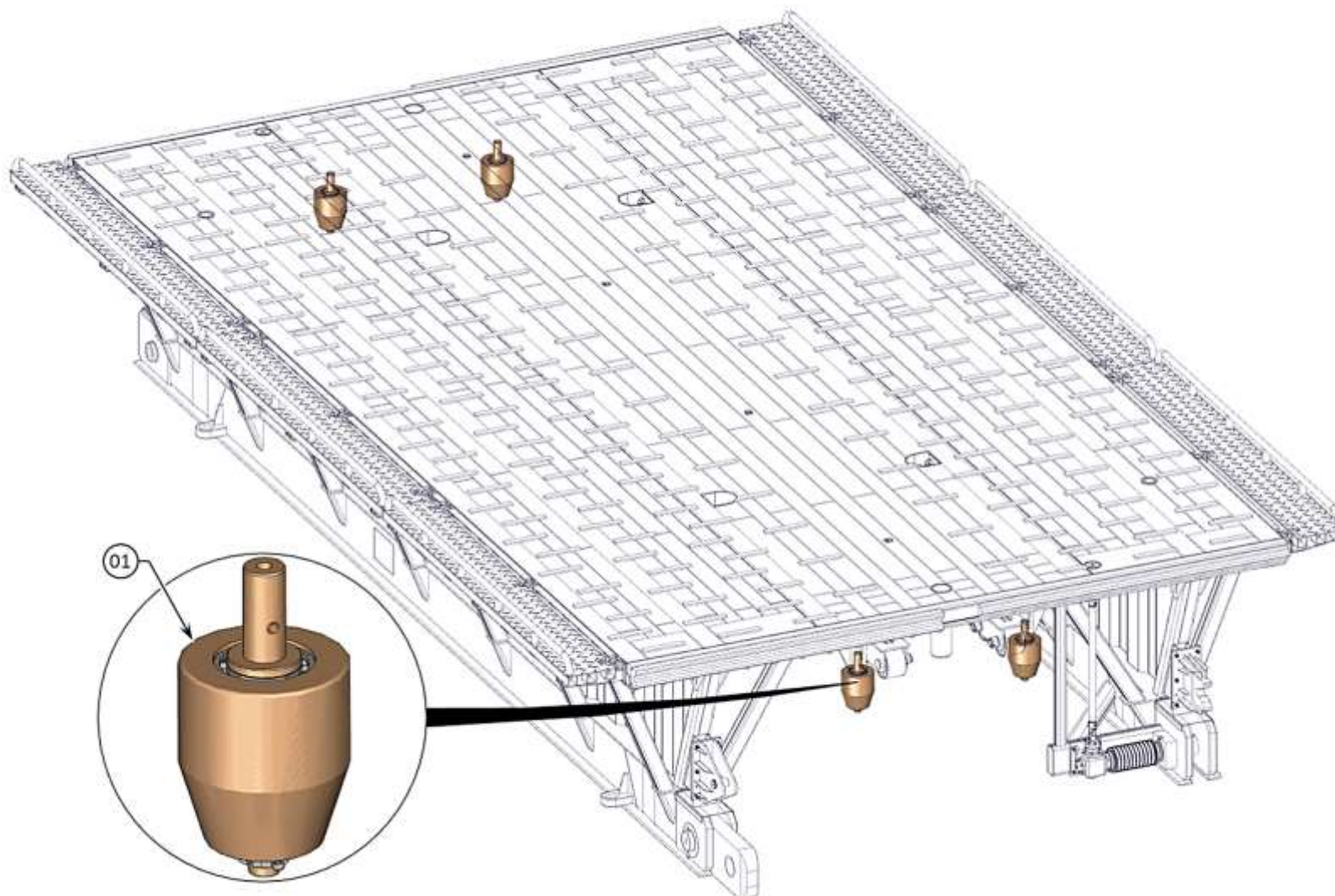
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
43.	01			BRACKET ASSY	1	410A215100000		8,199	
44.	02			SUPPORT ROLLER FIXING BRACKET	1	410A215200000		45,074	
45.	03			SUPPORT ROLLER	2	410A215000100		94,531	
46.	04			SPACER-A	2	410A215000200		18,104	
47.	05	P		PIN FOR BRACKET FIXING	1	410A215000300		14,130	
48.	06	P		PIN FOR BEARING	2	410A215000400		14,130	
49.	07			SLOTTED NUT M30 X 2 IS:2232-P-6	2	410A215000600		365	
50.	08			MACHINED WASHER 31-IS:2016-BRASS	2	410A215000700		18	
51.	09			SPLIT PIN IS 549-8x63-St	2	410A215000800		13	
52.	10			NEEDLE ROLLER BEARING-NKIS 35 (SIZE: 35x58x22) MAKE:INA	4	410A215000900		1,553	
53.	11			SPLIT PIN IS 549-5 x 36-St	1	410A215001000		9	

Note: The quantity mentioned in above table is for 1 support roller assembly (1 BCM contains 4 nos. of support roller assembly). 1 set of MB contains 5 nos. of BCM.

SUPPORT ROLLER ASSEMBLY (BCM) - VIEW 1

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PLATE - X



GUIDE ROLLER ASSEMBLY (BCM)

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(34)

RESTRICTED

PLATE - X

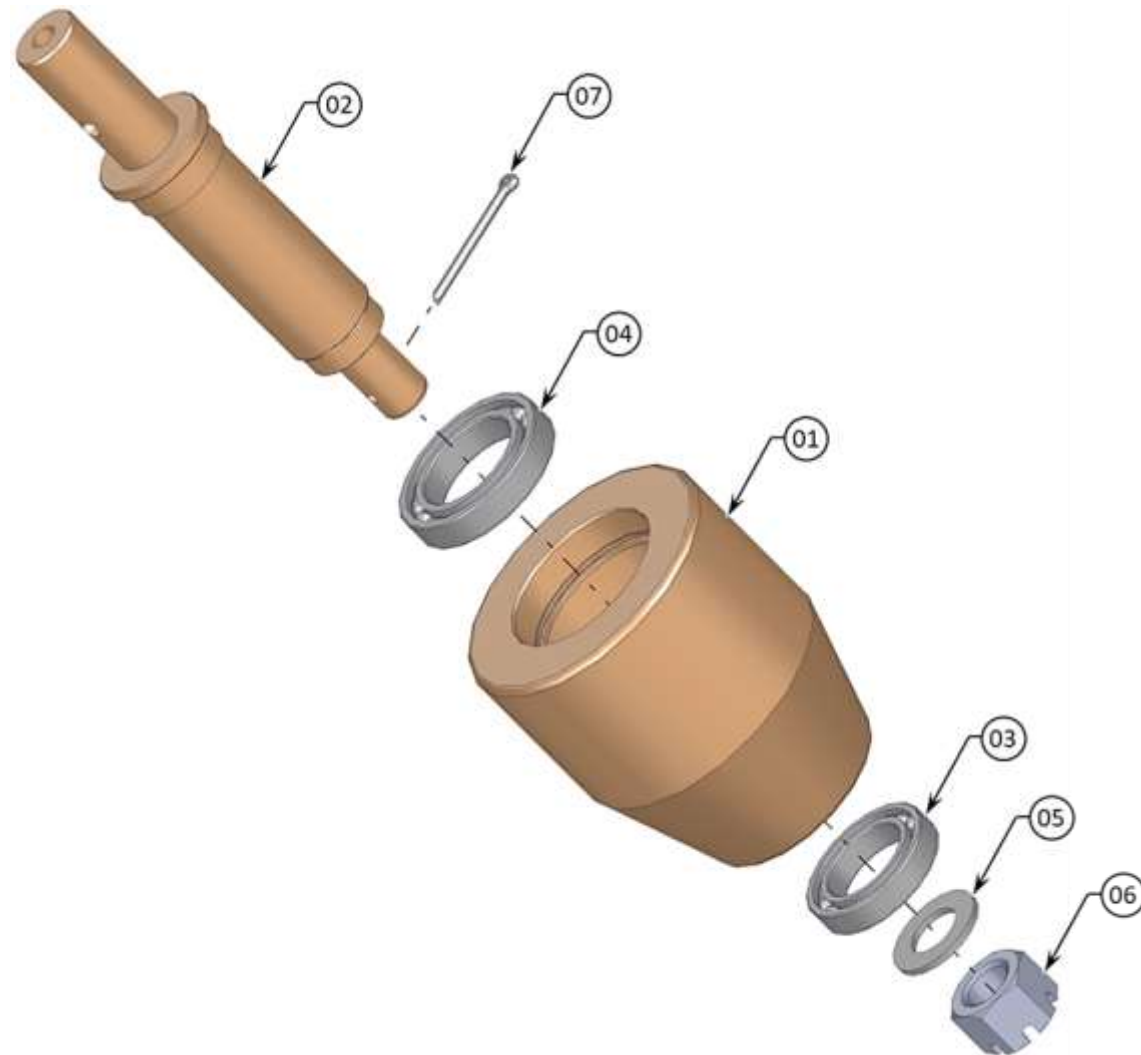
SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
54.	01			GUIDE ROLLER ASSEMBLY (BCM)	1	410A217000000		2,16,140	

Note: The quantity mentioned in above table is for 1 guide roller assembly (1 BCM contains 4 nos. of guide roller assembly). 1 set of MB contains 5 nos. of BCM.

GUIDE ROLLER ASSEMBLY (BCM)

RESTRICTED

RESTRICTED
PLATE - XI



GUIDE ROLLER ASSEMBLY (BCM) - VIEW 1

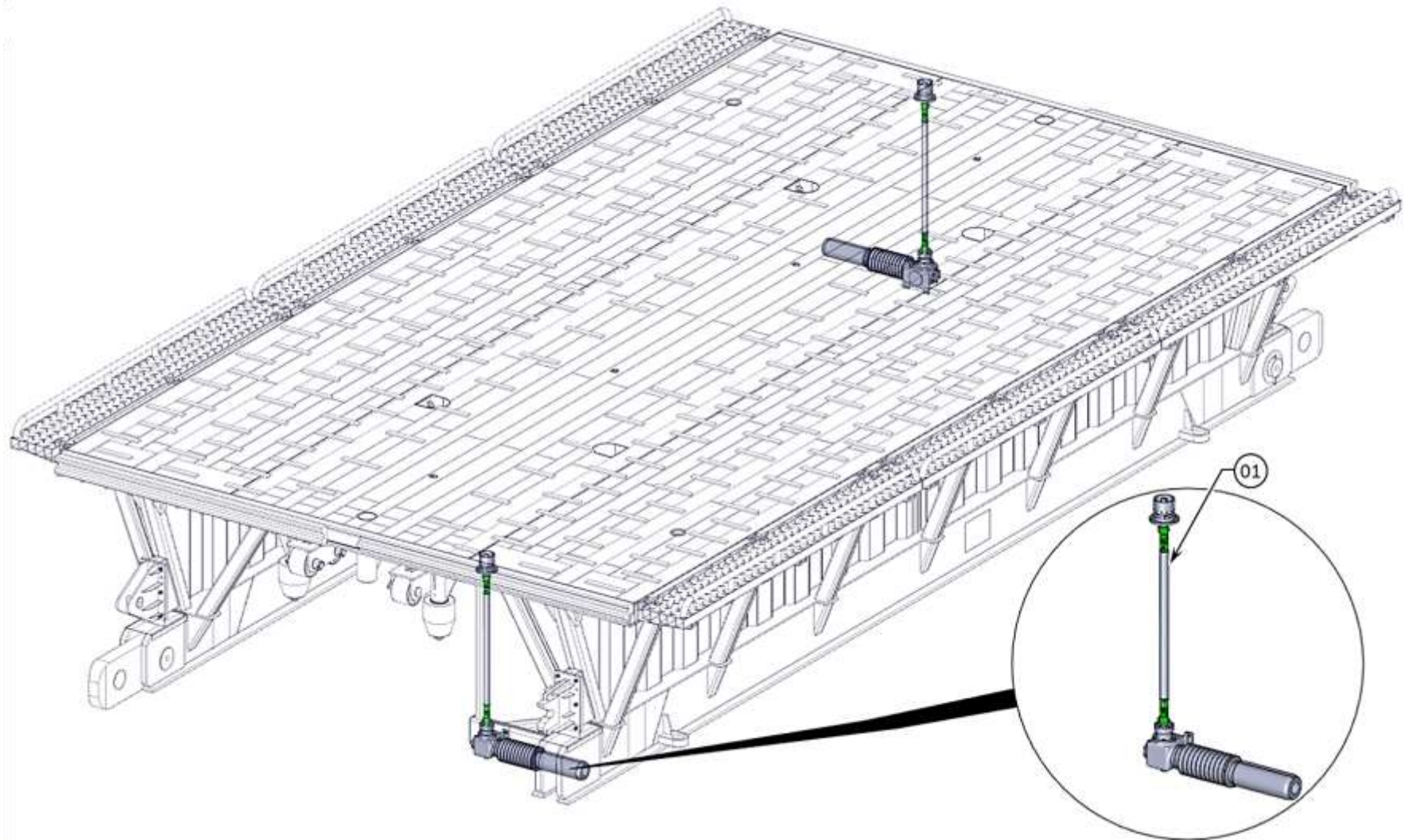
RESTRICTED
PLATE - XI

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
55.	01			GUIDE ROLLER	1	410A217000100		1,01,925	
56.	02			GUIDE ROLLER FIXING SHAFT	1	410A217000200		78,215	
57.	03			DEEP GROOVE BALL BEARING-SKF-6010-2RS1-(SIZE:50X80X16)	1	410A217000300		2,011	
58.	04			DEEP GROOVE BALL BEARING - SKF-6012-2RS1 (SIZE: 60X95X18)	1	410A217000400		2,192	
59.	05			MACHINED WASHER 31 - IS:2016-STEEL	1	410A217000500		18	
60.	06			SLOTTED NUT M30 x 2 IS:2232-P-6	1	410A217000600		365	
61.	07			SPLIT PIN IS 549-6.3 x 71-St	1	410A217000700		9	

Note: The quantity mentioned in above table is for 1 guide roller assembly (1 BCM contains 4 nos. of guide roller assembly). 1 set of MB contains 5 nos. of BCM.

GUIDE ROLLER ASSEMBLY (BCM) - VIEW 1

RESTRICTED
PLATE - XII



BRIDGE TO BRIDGE LOCKING MECHANISM (BCM)

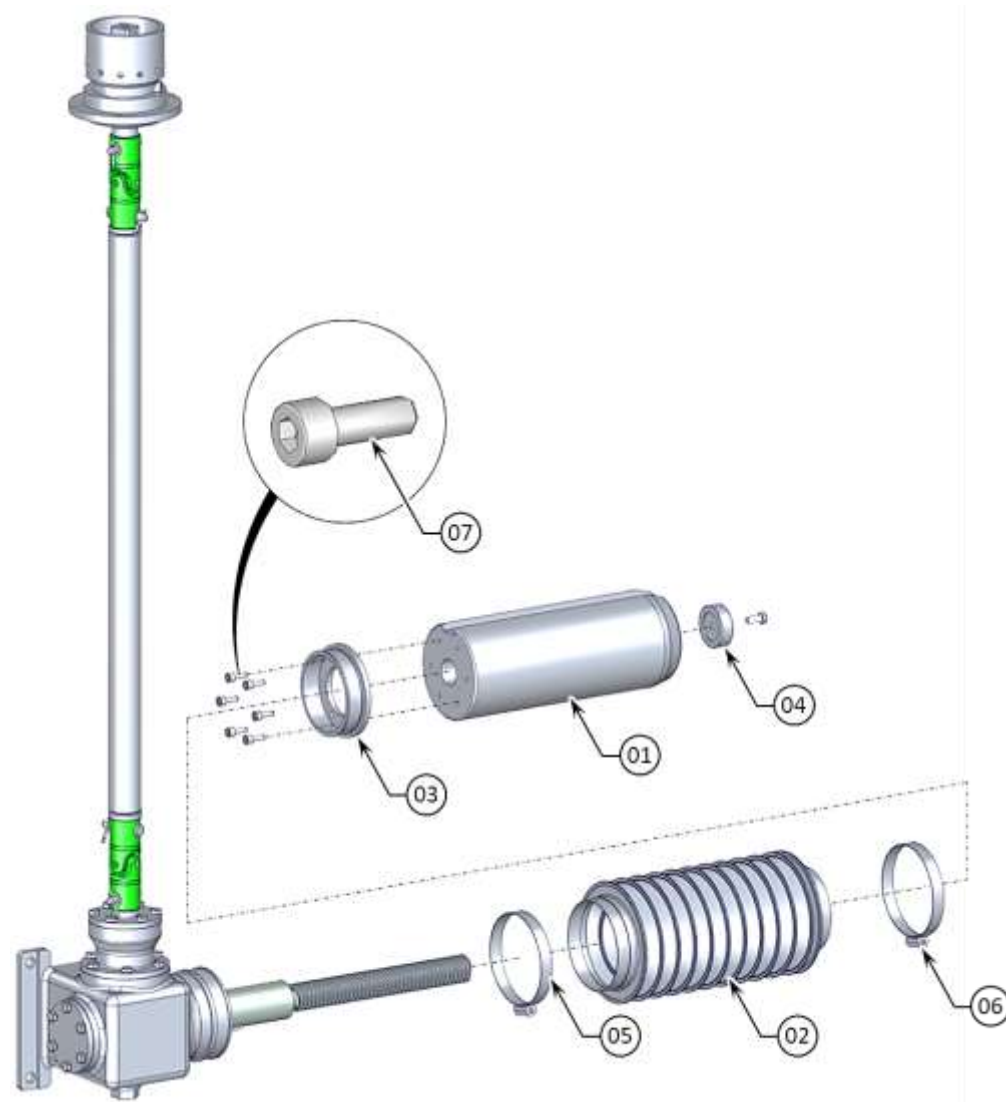
RESTRICTED
PLATE - XII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
62.	01			BRIDGE TO BRIDGE LOCKING MECHANISM (BCM)	1	410A218000000		5,93,759	

Note: The quantity mentioned in above table is for 1 bridge to bridge locking mechanism (1 BCM contains 2 nos. of bridge to bridge locking mechanism). 1 set of MB contains 5 nos. of BCM.

BRIDGE TO BRIDGE LOCKING MECHANISM (BCM)

RESTRICTED
PLATE - XIII



BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 1

RESTRICTED

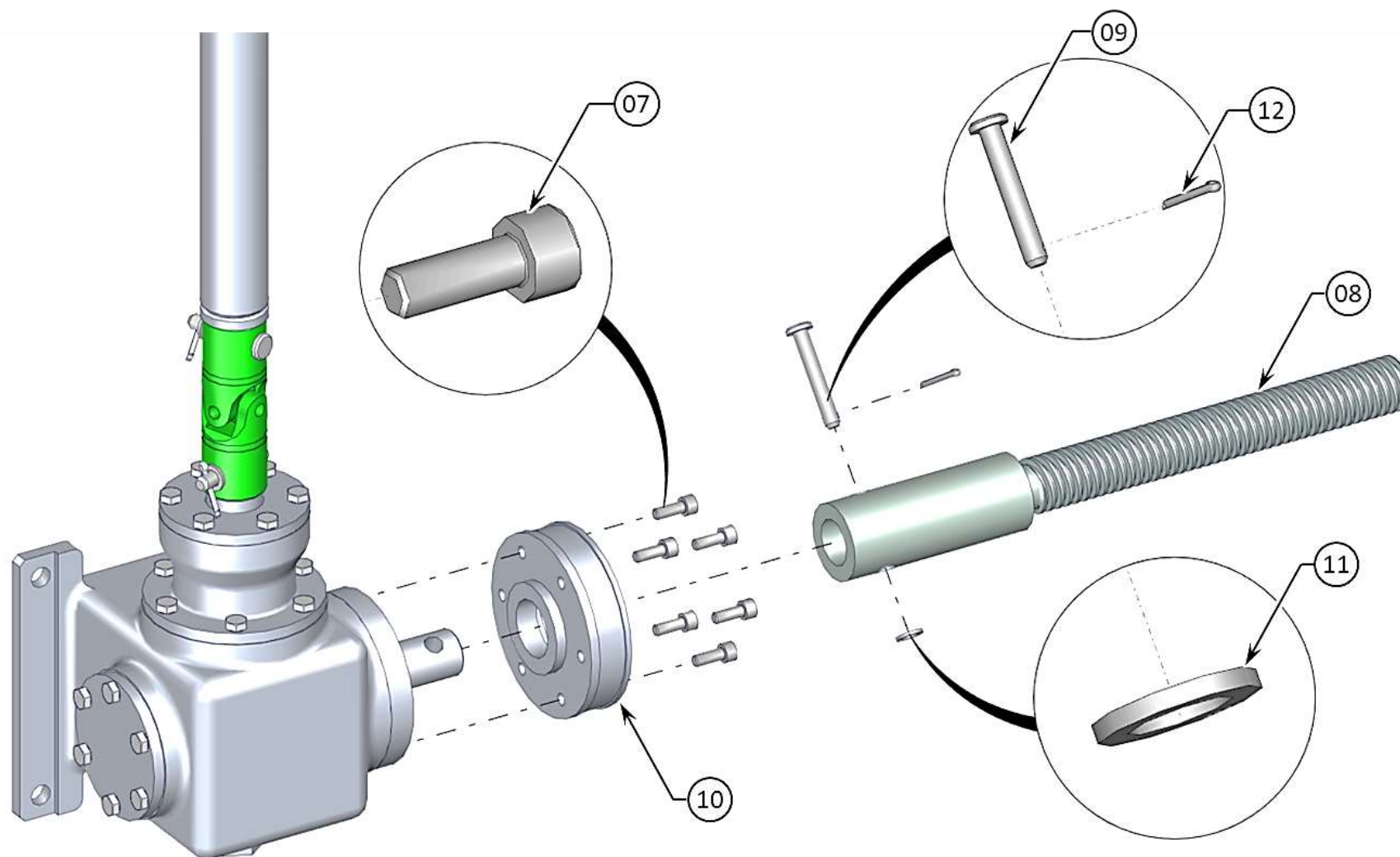
RESTRICTED
PLATE - XIII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
63.	01	P		BRIDGE TO BRIDGE LOCKING PIN	1	410A31D000100		2,06,980	
64.	02	P		BELLOW	1	410A31D000300		16,820	
65.	03			BELLOW MOUNTING COVER	1	410A31D000600		1,828	
66.	04	P		CAP	1	410A22D000800		5,180	
67.	05			WORM CLIP (HEAVY DUTY)	1	410A22D001300		1,828	
68.	06			WORM CLIP (HEAVY DUTY)	1	410A218000E00		1,828	
69.	07			HEX. SOC. HEAD CAP SCREW ISO 4014-M6x16-10.9	6	410A218000F00		1,828	

Note: The quantity mentioned in above table is for 1 bridge to bridge locking mechanism (1 BCM contains 2 nos. of bridge to bridge locking mechanism). 1 set of MB contains 5 nos. of BCM.

BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 1

RESTRICTED
PLATE - XIII (CONTINUED)



BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 2

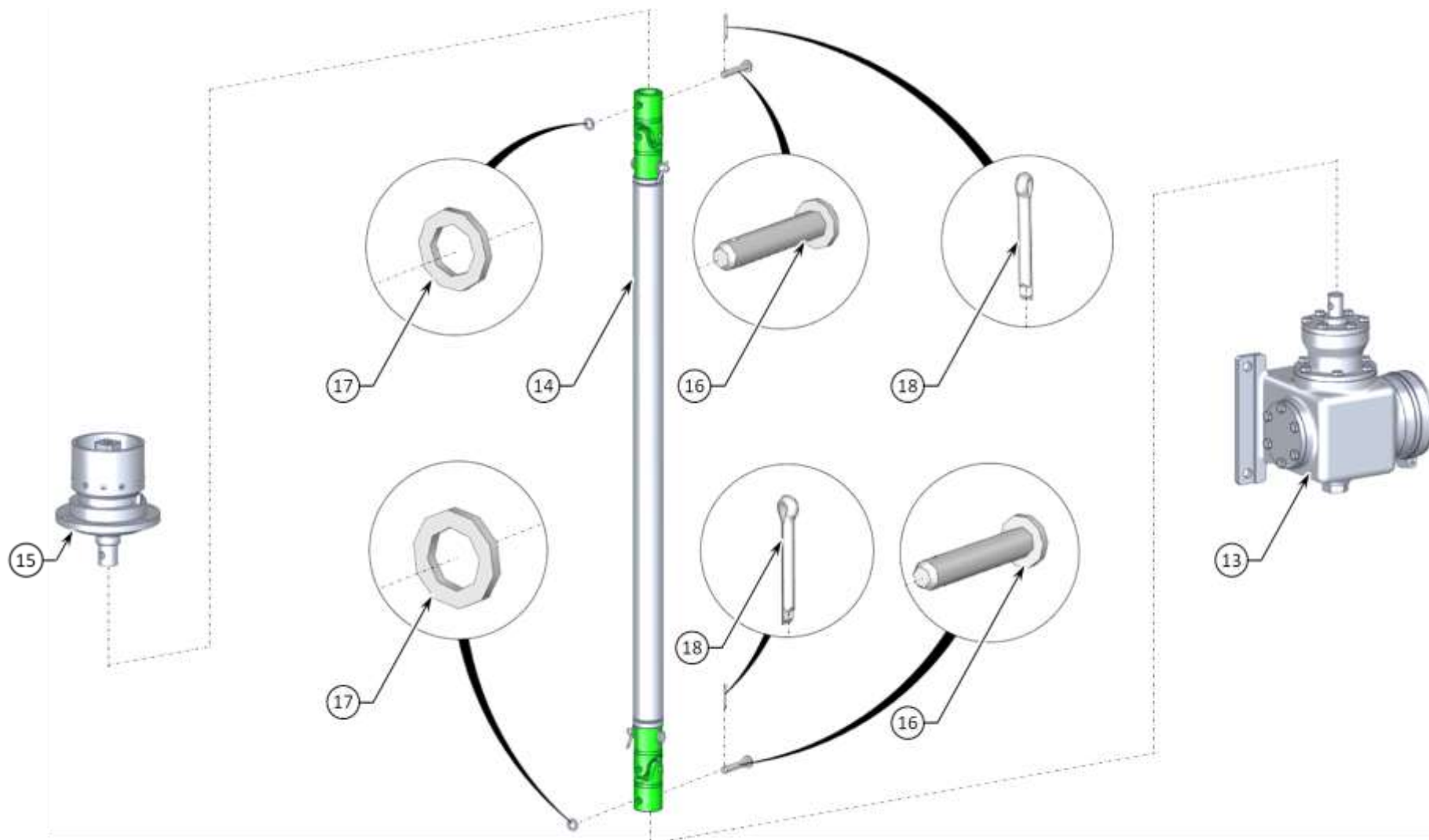
RESTRICTED
PLATE - XIII (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
70.	07			HEX. SOC. HEAD CAP SCREW ISO 4014-M6x16-10.9	6	410A218000F00		1,828	
71.	08			SCREW	1	410A31D000200		914	
72.	09			CLEVIS PIN-1	1	410A31D000400		2,742	
73.	10			BEARING COVER	1	410A22D000700		1,828	
74.	11			MACHINED WASHER 8.4 IS:2016-STEEL	1	410A22D001100		1,828	
75.	12			SPLIT PIN ISO 1234-2x12-St	1	410A22D001200		1,828	

Note: The quantity mentioned in above table is for 1 bridge to bridge locking mechanism (1 BCM contains 2 nos. of bridge to bridge locking mechanism). 1 set of MB contains 5 nos. of BCM.

BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 2

RESTRICTED
PLATE - XIII (CONTINUED)



BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 3

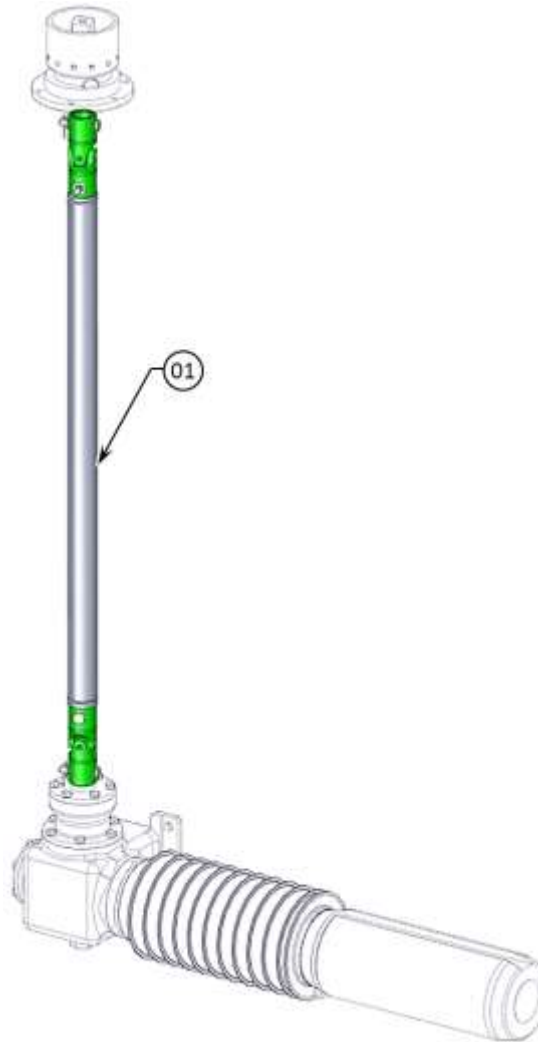
RESTRICTED
PLATE - XIII (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
76.	13			ASSEMBLY BEVEL GEARBOX	1	410A218100000		1,58,996	
77.	14			ASSEMBLY CONNECTING SHAFT	1	410A218200000		57,732	
78.	15			ASSY CONNECTING SHAFT WITH BEARING	1	410A218300000		18,177	
79.	16			CLEVIS PIN-II	2	410A31D000500		4,570	
80.	17			MACHINED WASHER 10.5 IS:2016-STEEL	2	410A22D000900		1,828	
81.	18			SPLIT PIN ISO 1234-2x12-St	2	410A22D001000		1,828	

Note: The quantity mentioned in above table is for 1 bridge to bridge locking mechanism (1 BCM contains 2 nos. of bridge to bridge locking mechanism). 1 set of MB contains 5 nos. of BCM.

BRIDGE TO BRIDGE LOCKING MECHANISM (BCM) - VIEW 3

RESTRICTED
PLATE - XIV



ASSEMBLY CONNECTING SHAFT (BRIDGE TO BRIDGE LOCKING MECHANISM)

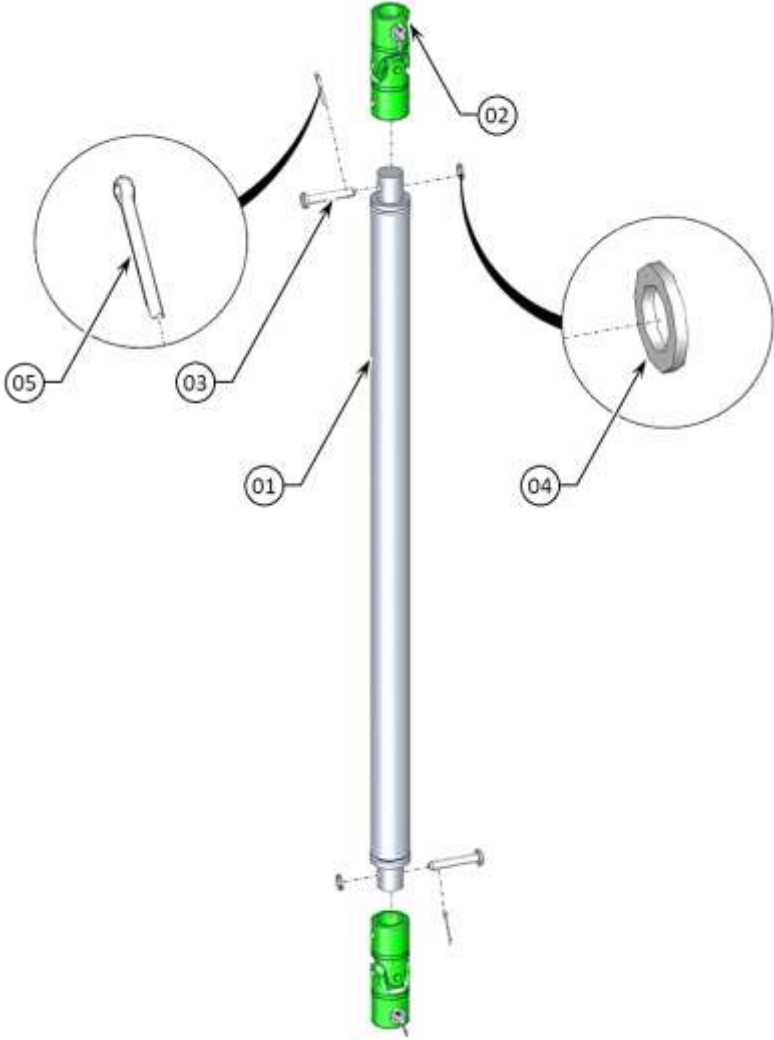
RESTRICTED
PLATE - XIV

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
82.	01			ASSEMBLY CONNECTING SHAFT (BRIDGE TO BRIDGE LOCKING MECHANISM)	1	410A218200000		57,732	

Note: The quantity mentioned in above table is for 1 assembly connecting shaft (bridge to bridge locking mechanism). 1 BCM contains 2 nos. of bridge to bridge locking mechanism. 1 set of MB contains 5 nos. of BCM.

ASSEMBLY CONNECTING SHAFT (BRIDGE TO BRIDGE LOCKING MECHANISM)

PLATE - XV



ASSEMBLY CONNECTING SHAFT (BRIDGE TO BRIDGE LOCKING MECHANISM) - VIEW 1

RESTRICTED

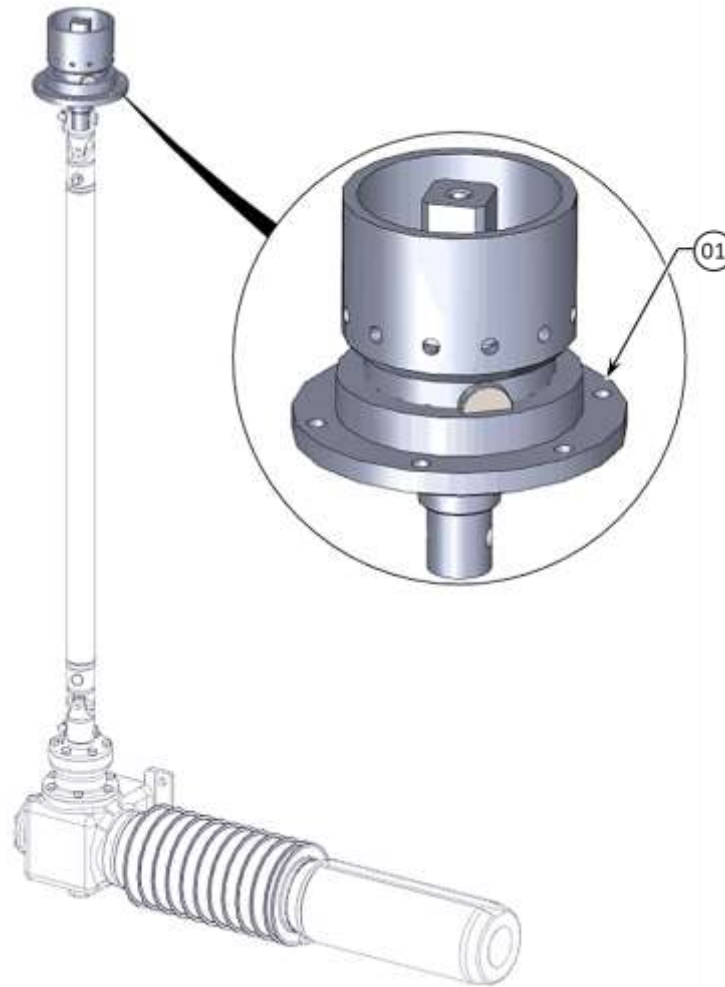
RESTRICTED
PLATE - XV

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
83.	01			TUBE ASSY	1		410A218210000	27,413	
84.	02	P		ASSEMBLY UNIVERSAL JOINT	2		410A218220000	21,030	
85.	03			CLEVIS PIN-II	4		410A31D000500	18,275	
86.	04			MACHINED WASHER 8.4-IS:2016-STEEL	4		410A22D000900	1,828	
87.	05			SPLIT PIN ISO 1234-2x12-St	4		410A22D001000	1,828	

Note: The quantity mentioned in above table is for 1 assembly connecting shaft (bridge to bridge locking mechanism). 1 BCM contains 2 nos. of bridge to bridge locking mechanism. 1 set of MB contains 5 nos. of BCM.

ASSEMBLY CONNECTING SHAFT (BRIDGE TO BRIDGE LOCKING MECHANISM) - VIEW 1

RESTRICTED
PLATE - XVI



ASSY CONNECTING SHAFT WITH BEARING (BRIDGE TO BRIDGE LOCKING MECHANISM)

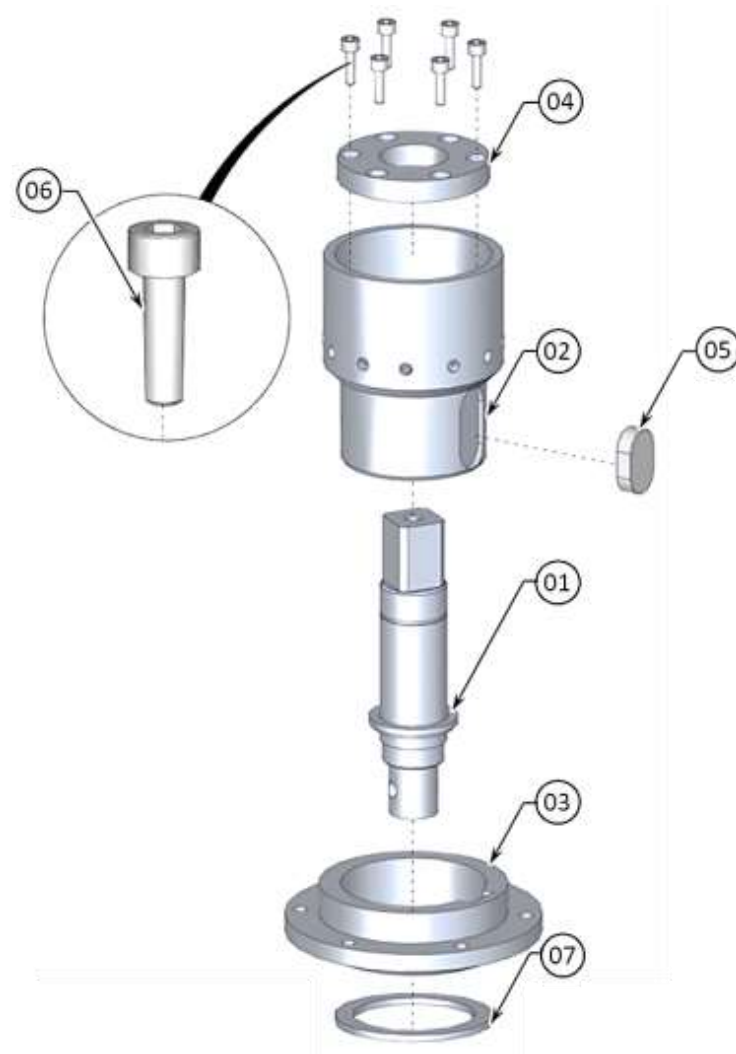
RESTRICTED
PLATE - XVI

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
88.	01			ASSY CONNECTING SHAFT WITH BEARING (BRIDGE TO BRIDGE LOCKING MECHANISM)	1	410A218300000		18,177	

Note: The quantity mentioned in above table is for 1 assy connecting shaft with bearing (bridge to bridge locking mechanism). 1 BCM contains 2 nos. of bridge to bridge locking mechanism. 1 set of MB contains 5 nos. of BCM.

ASSY CONNECTING SHAFT WITH BEARING (BRIDGE TO BRIDGE LOCKING MECHANISM)

RESTRICTED
PLATE - XVII



ASSY CONNECTING SHAFT WITH BEARING (BRIDGE TO BRIDGE LOCKING MECHANISM) - VIEW 1

RESTRICTED
PLATE - XVII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
89.	01			SHAFT	1		410A218300100	2,742	
90.	02			INPUT SHAFT HOUSING	1		410A218300200	2,742	
91.	03			BEARING HOUSING SLEEVE	1		410A218300300	1,828	
92.	04			SEALING COVER	1		410A218300400	1,828	
93.	05			PARALLEL KEY BOTH END ROUND	1		410A218300700	2,742	
94.	06			HEX. SOC. HEAD CAP SCREW	7		410A218300900	1,828	
95.	07			EXTERNAL CIRCLIP	2		410A218301000	1,828	

Note: The quantity mentioned in above table is for 1 assy connecting shaft with bearing (bridge to bridge locking mechanism). 1 BCM contains 2 nos. of bridge to bridge locking mechanism. 1 set of MB contains 5 nos. of BCM.

ASSY CONNECTING SHAFT WITH BEARING (BRIDGE TO BRIDGE LOCKING MECHANISM) - VIEW 1

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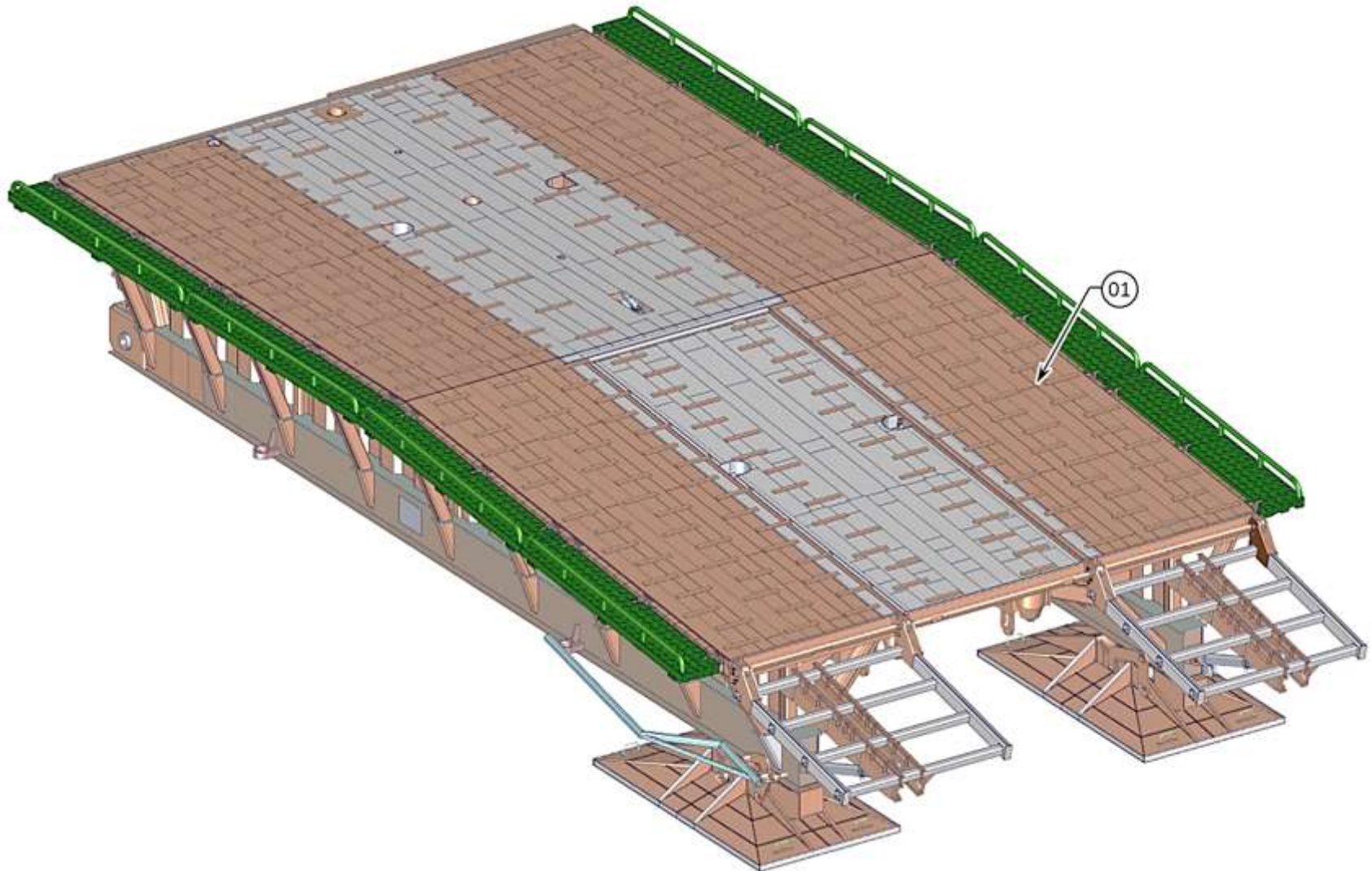
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SUB - SECTION - 2

BRIDGE END MODULE ASSEMBLY

RESTRICTED

RESTRICTED
PLATE - XVIII



BRIDGE END MODULE ASSY

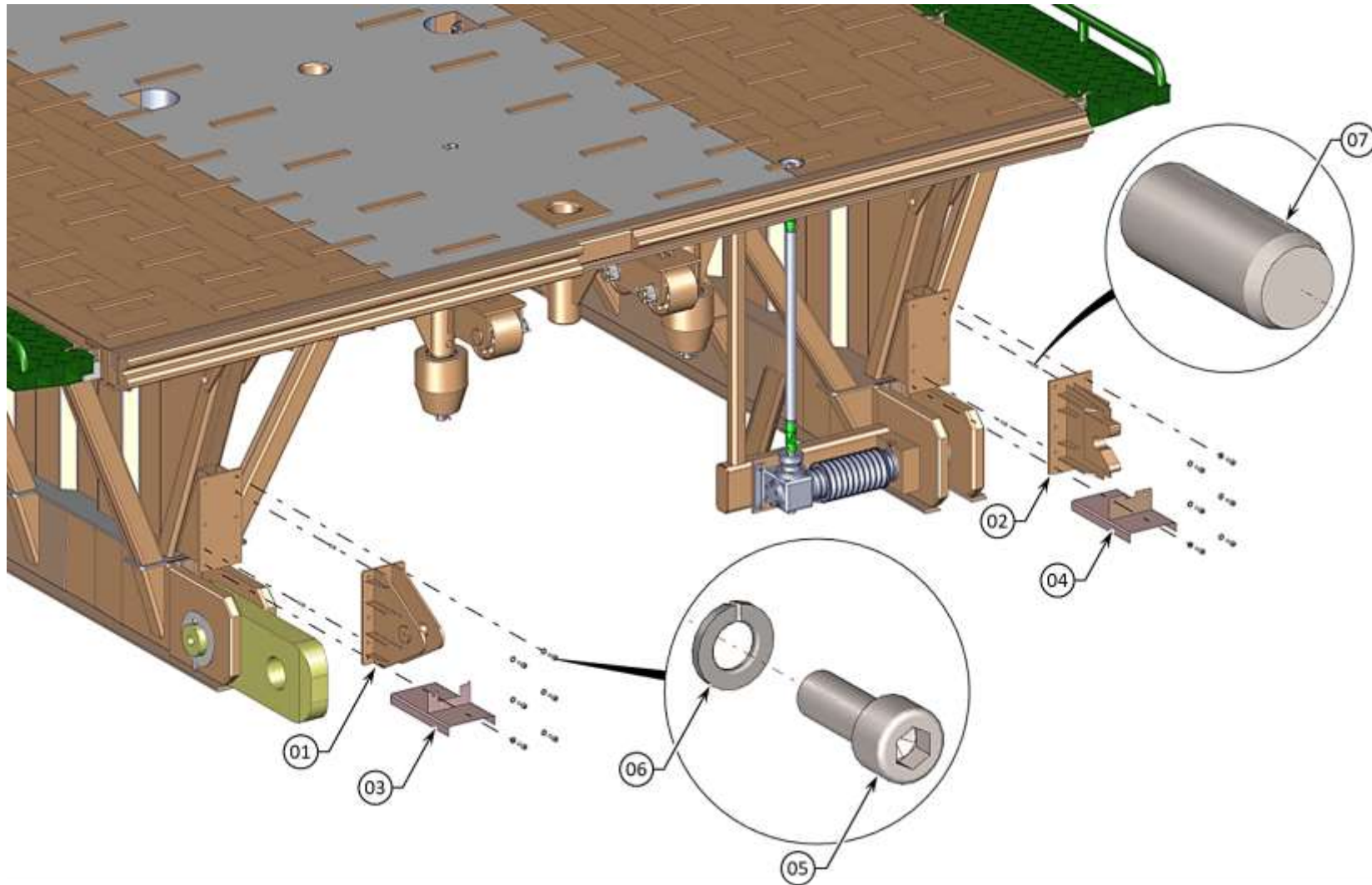
RESTRICTED
PLATE - XVIII

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
96.	01			BRIDGE END MODULE ASSY	1	410A220000000		2,41,77,017	

Note: The quantity mentioned in above table is for 1 BEM. 1 set of MB contains 4 nos. of BEM.

BRIDGE END MODULE ASSY

RESTRICTED
PLATE - XIX



BRIDGE END MODULE ASSY - VIEW 1

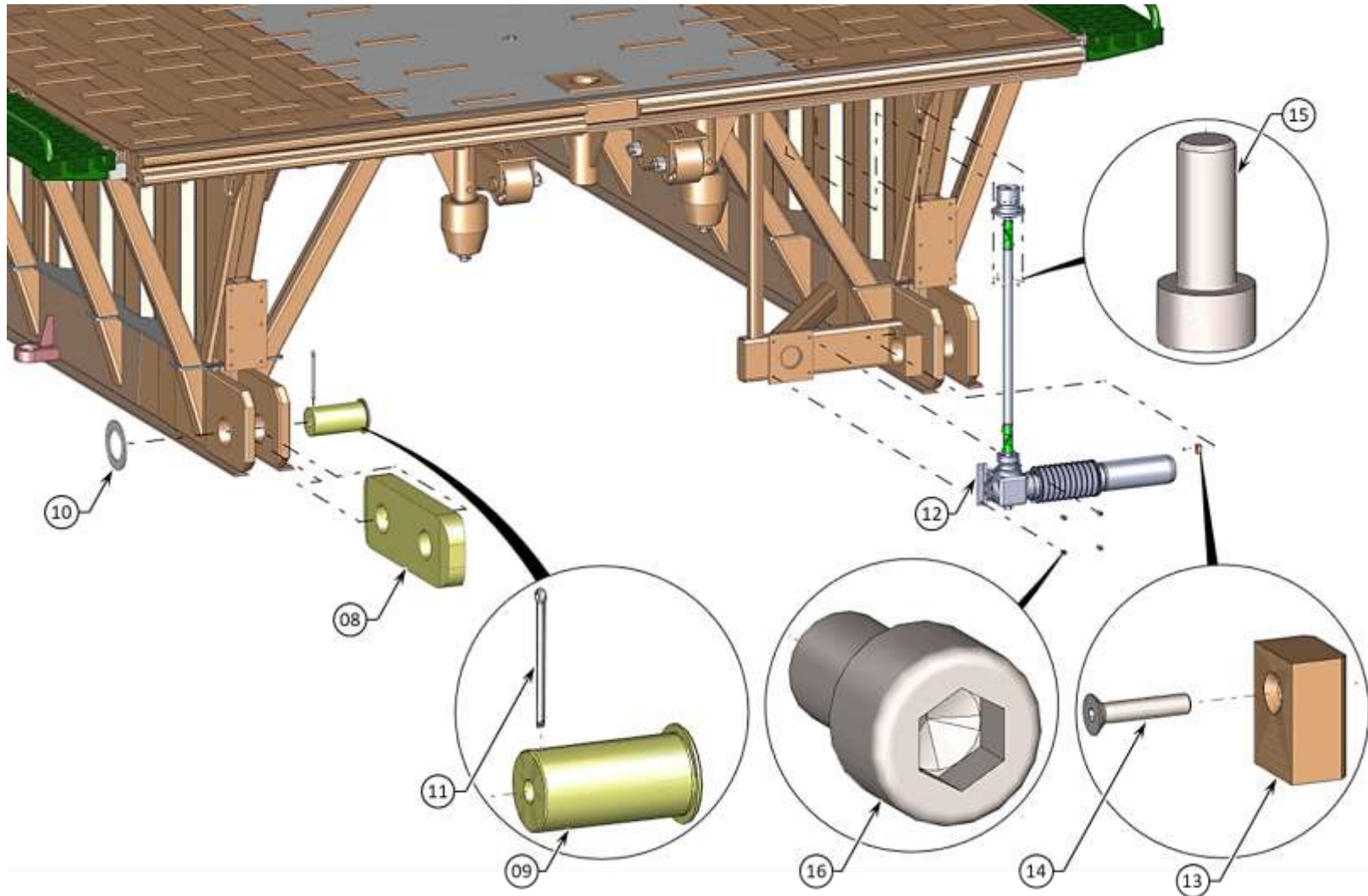
RESTRICTED
PLATE - XIX

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
97.	01			GUIDE BEAK MALE ASSEMBLY	1	410A213000000		84,557	
98.	02			GUIDE BEAK FEMALE ASSY	1	410A214000000		50,978	
99.	03			COVER ASSY-1	1	410A21A000000		2,988	
100.	04			COVER ASSY-2	1	410A21B000000		2,988	
101.	05			HEX. SOCKET HEAD CAP SCREW ISO 4762-M10x25-12.9	12	410A210001700		50	
102.	06			SPRING LOCK WASHER B10- IS 3063 -STEEL	12	410A210002000		18	
103.	07			PARALLEL PIN IS 2393 10h8 X 26 - St	4	410A210002500		27	

Note: The quantity mentioned in above table is for 1 BEM. 1 set of MB contains 4 nos. of BEM.

BRIDGE END MODULE ASSY - VIEW 1

RESTRICTED
PLATE - XIX (CONTINUED)



BRIDGE END MODULE ASSY - VIEW 2

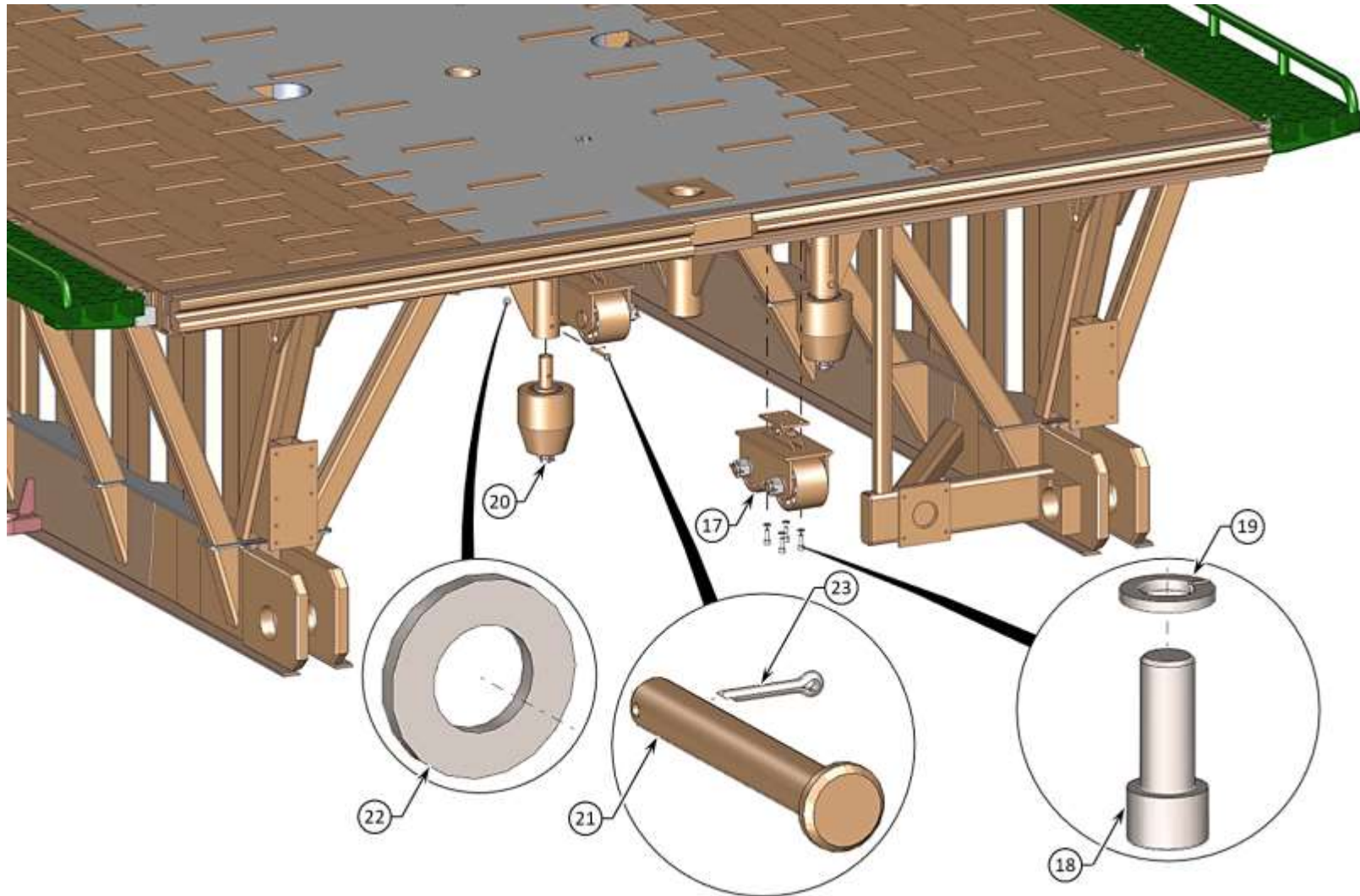
RESTRICTED
PLATE - XIX (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
104.	08			LINK PLATE	1	410A210000400		1,43,683	
105.	09			BRIDGE LOCKING PIN	1	410A210000600		1,07,324	
106.	10			SPACER	1	410A210001000		2,742	
107.	11			SPLIT PIN IS 549-10 x 125-St	1	410A210002300		89	
108.	12			BRIDGE TO BRIDGE LOCKING MECHANISM	1	410A218000000		5,93,759	
109.	13			TENON	1	410A210000900		2,376	
110.	14			HEXAGON SOCKET COUNTERSUNK HEAD SCREW M5x25 IS 6761	1	410A210001400		9	
111.	15			HEX.SOCKET HEAD CAP SCREW ISO 4762-M6x16-12.9	10	410A210001500		18	
112.	16			HEX.SOCKET HEAD CAP SCREW ISO 4762-M10x16-12.9	4	410A210001600		27	

Note: The quantity mentioned in above table is for 1 BEM. 1 set of MB contains 4 nos. of BEM.

BRIDGE END MODULE ASSY - VIEW 2

RESTRICTED
PLATE - XIX (CONTINUED)



BRIDGE END MODULE ASSY - VIEW 3

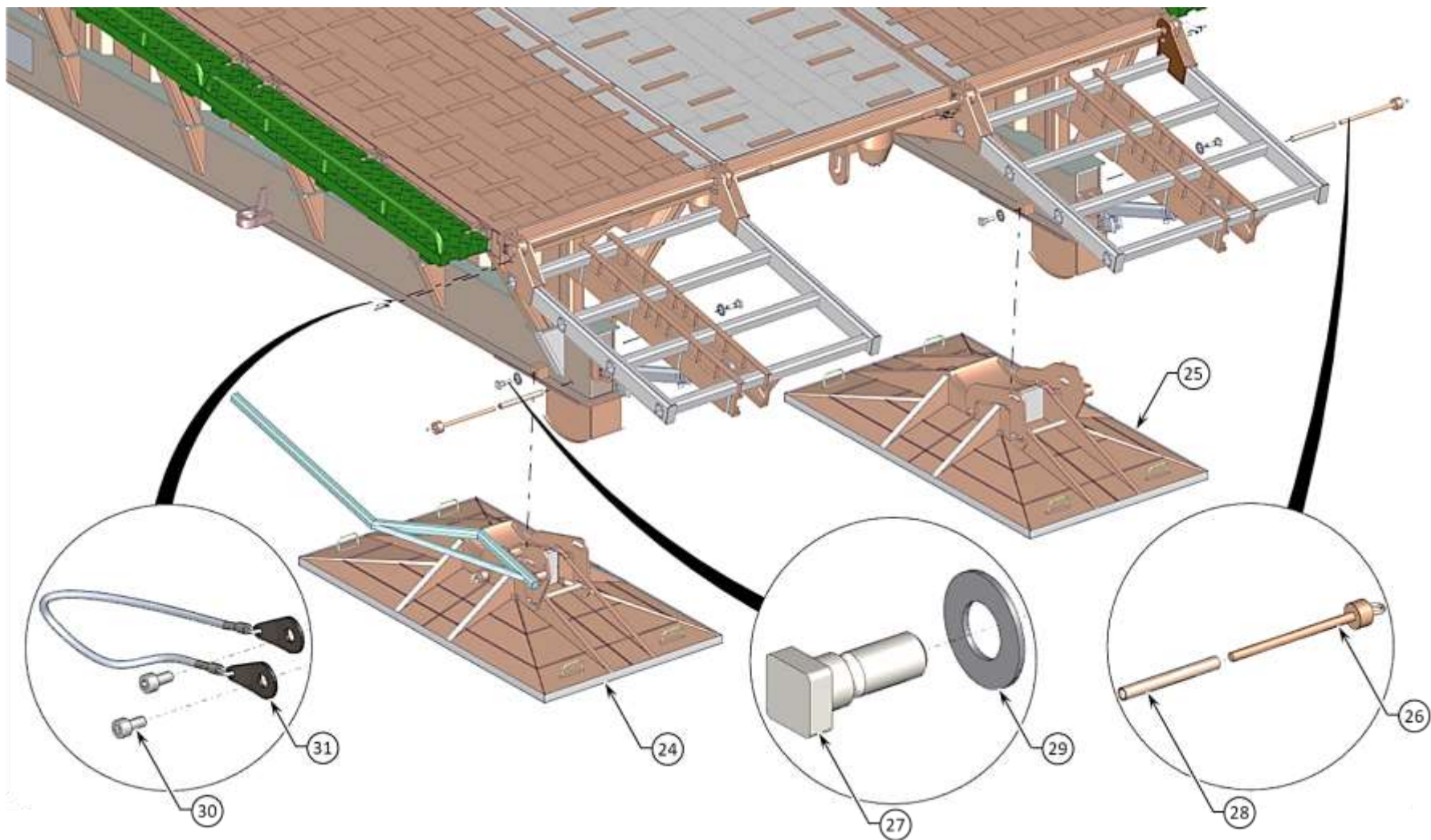
RESTRICTED
PLATE - XIX (CONTINUED)

SR. NO.	PLATE REF / ITEM NO.	'P' PROVIDE	NSN	DESIGNATION	NO. OFF	DRG. NO.	FIRMS PART NO.	UNIT PRICE RS.	REMARK
113.	17			SUPPORT ROLLER ASSEMBLY	2	410A215000000		3,34,100	
114.	18			HEX.SOCKET HEAD CAP SCREW ISO 4762-M12x30-12.9	8	410A210001800		40	
115.	19			SPRING LOCK WASHER B12- IS 3063 -STEEL	8	410A210002100		18	
116.	20			GUIDE ROLLER ASSEMBLY	2	410A217000000		2,16,140	
117.	21			LOCKING PIN FOR GUIDE ROLLER ASSY	2	410A210000700		35,766	
118.	22			MACHINED WASHER 13 - IS : 2016 STEEL	2	410A210001900		9	
119.	23			SPLIT PIN IS 549-3.2 x 16-St	2	410A210002200		18	

Note: The quantity mentioned in above table is for 1 BEM. 1 set of MB contains 4 nos. of BEM.

BRIDGE END MODULE ASSY - VIEW 3

RESTRICTED
PLATE - XIX (CONTINUED)



BRIDGE END MODULE ASSY - VIEW 4